

A TO Z DEVELOPMENT GUIDE

LAM – TV Request Management System

How to build the TVRMS on Microsoft Power Platform, step by step, with sample data and use cases.

Based on	ABC_REQUEST.pptx – HCLTech Solution Approach, September 2026
Platform	Power Apps (Canvas + Model-Driven), Power Automate, Dataverse, SharePoint Online, Outlook, Power BI
Prepared	October 2026
Contents	16 guide documents + 10 sample data files

Items marked **(ASSUMPTION)** are not in the PPT and must be confirmed with the business during the "Plan for Success" phase.

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Tip: use the bookmarks panel of your PDF reader to jump to any heading. All links in this PDF are clickable.

RADIANCE – LAM TV (Test Vehicle) Request Management System

This repository has a **complete A to Z guide** for building the "**LAM – TV Request Management System**" (TVRMS).

The only input given by the manager was the PowerPoint file **ABC_REQUEST.pptx** (HCLTech's solution approach, September 2026). Everything here is based on that file. Where the file does not give a detail, we made a **sensible assumption**. Every assumption is marked with **(ASSUMPTION)** so that you can confirm it with your manager.

1. What is this project? (in one paragraph)

Today, LAM manages **Test Vehicle (TV) requests** with emails, PowerPoint files, Excel sheets and loose documents. Nobody can see the full picture in one place. We will build **one central system** on **Microsoft Power Platform** where a request is **created → reviewed → assessed (feasibility) → funded → executed → closed**, with all documents stored in **SharePoint**, all emails sent automatically by **Power Automate**, and management reports in **Power BI**.

2. Technology used (from slide "Solution Architecture")

Layer	Tool	What it does
Login / Security	Microsoft Entra ID	Users sign in with their company account
UI for end users	Power Apps Canvas App	Request intake, review, feasibility, funding, execution, closure
UI for admins	Power Apps Model-Driven App	Master data, notification logs, audit
Automation	Power Automate	Emails, approvals, reminders, folder creation, PDF
Data	Dataverse	All request data, security roles, audit
Documents	SharePoint Online document library	One folder (workspace) per TVID
Emails	Outlook (Office 365)	All notifications
Reports	Power BI	Status, cycle time, aging, cost, utilization

3. How to read this guide (follow in order)

#	File	What you will learn
0	docs/00-Is-The-Documents-Enough	Is the PPT enough? Gaps + list of questions to ask your manager
1	docs/01-Project-Understanding	Slide-by-slide explanation, roles, glossary, 6 phases
2	docs/02-Environment-Setup	Licenses, environments, solution, publisher, SharePoint site
3	docs/03-Dataverse-Data-Model	All tables, columns, choices, relationships, with sample data
4	docs/04-SharePoint-Documents-Management	TVID workspace, folders, versioning, permissions, 7-year retention
5	docs/05-Security-Roles	Role-based access, row-level security
6	docs/06-Canvas-App	Every screen, with Power Fx formulas
7	docs/07-Model-Driven-App	Admin app for master data and logs
8	docs/08-Power-Automate-Flows	All flows step by step (notifications, approvals, reminders, PDF)

#	File	What you will learn
9	docs/09-Power-BI-Reports	Dashboard pages and DAX measures
10	docs/10-Testing-and-UAT	Test cases with sample data
11	docs/11-Deployment-GoLive-Hypercare	Moving Dev → Test → Prod, go-live checklist, hypercare
12	docs/12-Project-Plan-9-Weeks	Week-by-week plan, sprint backlog
13	docs/13-End-to-End-Use-Case	One full request story from start to end (with sample data)
–	sample-data/	Ready CSV files to import into Dataverse for testing

4. Quick answer: "Is this document enough to develop the project?"

Short answer: No, not fully. The PPT is a **proposal / solution approach** (what will be built, the timeline and the price). It is **good enough to start the "Plan for Success" phase (Week 1–2)** and to set up the platform, but it is **not a detailed requirement document**. Field lists, status rules, reminder days, approval rules, email texts, report KPIs and master data are missing. These are exactly the things the PPT says will be finalised in Week 1–2 ("Solution Blueprint"). See [docs/00-Is-The-Document-Enough](#) for the full gap list and the questions to ask.

00 – Is the PPT document enough to develop the project?

1. What kind of document is `ABC_REQUEST.pptx` ?

It is a "**Solution Approach**" / **proposal** made by HCLTech for the customer **LAM**. It has 10 slides:

Slide	Title	What it gives us	Detail level
1	Cover – LAM TV Request Management System	Project name, date (Sep 2026)	–
2	Table of Contents	7 sections	–
3	Objectives & Scope	Problem, goal, 13 in-scope items, 4 out-of-scope items	Medium
4	Business Process Flow	6 phases, user journey, key actions, notifications	Medium
5	Solution Architecture	Tools: Canvas app, Model-driven app, Power Automate, Dataverse, SharePoint, Outlook, Power BI	Medium
6	Execution Approach	Plan → Build (2 sprints) → Deploy → Hypercare	High-level
7	Execution Timeline	9 weeks plan	High-level
8	Assumptions & Dependencies	Customer gives master data, licenses, SMEs	High-level
9	Non-Functional Requirements	24x7, Edge, SharePoint, 7-year retention, audit, RBAC, scalable	Medium
10	Commercials	Fixed price USD 58,950 (excluding taxes)	Not for developers

2. Final answer

The document is enough to START the project, but NOT enough to FINISH building it.

- **✓ Enough for:** understanding the goal, choosing the technology, drawing the architecture, creating environments, creating a first draft of the data model, planning the 9 weeks, and preparing questions for the workshops.
- **✗ Not enough for:** final form fields, status rules, approval rules, email texts, reminder timings, PDF layout, report KPIs, and test cases with real values.

This is **normal**. Slide 6 and 7 clearly say that **Week 1–2 ("Plan for Success")** is for "Requirement Understanding" and "Solution Blueprint". So your **first job** is to collect the missing details below and write a **Solution Blueprint** (also called a design document). In this repository we filled each gap with a reasonable **(ASSUMPTION)** so you can build a working version now and adjust later.

3. What is clear in the document (we can build this directly)

1. 6 phases: Request Initiation → APPG Review → Internal Alignment → Funding Approval → Execution → Closure.
2. Request types: **New, Repeat, One-time, Recurring**.
3. Mandatory fields and **technical document** must be validated on submit.
4. APPG can **review, assign owner, comment, ask for more information, reject**.
5. Feasibility = compare **multiple Research Institutions / sourcing options** on capability, cost, timeline, constraints, recommendation.
6. Two internal alignment steps (1st: quotes/estimates; 2nd: confirm selected option & raise funding request).

7. Funding: find **Cost Center Owner**, route approval, capture decision + comments + history.
8. Execution: milestones, planned & actual dates, progress, risks, costs, comments, expected completion date.
9. Closure: deliverables, outcomes, lessons learned, costs, reuse recommendations.
10. SharePoint: auto-create a **TVID workspace** with fixed folders, versioning, permissions, **7-year retention**.
11. Automation: notifications, reminders, escalations, status updates, **automatic cancellation**, approvals, audit log, PDF generation.
12. Roles: Requester, APPG user, Cost Center Owner, Agreed stakeholders, System admin.
13. Power BI: Request status, cycle time & aging, cost tracking, utilization & performance, exception analysis, executive view.
14. Out of scope: integration with external systems, old data migration, user training, change management.

4. Gaps – what is missing (and what we assumed)

#	Missing item	Why it matters	Our ASSUMPTION in this guide
G1	Full list of fields on the request form	Needed to build table + screen	We designed ~25 fields (see doc 03)
G2	Meaning of "APPG"	Team name in the process	A central planning/governance team that reviews TV requests
G3	Exact status list and allowed moves	Drives the whole workflow	15 statuses, see doc 03 §4
G4	Which documents are "mandatory technical documents"	Validation on submit	At least 1 file of type "Technical Specification"
G5	Reminder / escalation / auto-cancel days	Scheduled flows	Reminder after 3 working days, escalation after 5 , auto-cancel after 30 days in "More Info Required" (stored in a Settings table so business can change)
G6	Funding approval rules (one level or many? amount limits?)	Approval flow	One approver = Cost Center Owner. If amount > USD 50,000 , also the CC Owner's manager (configurable)
G7	How is the Cost Center Owner found?	Routing	From the Cost Center master table (Owner column)
G8	Who are "agreed stakeholders"?	Security	Read-only users added per request (Access Team)
G9	SharePoint folder names	Folder creation flow	01-Request, 02-Technical Documents, 03-Feasibility & Quotes, 04-Funding Approval, 05-Execution, 06-Final Report & Closure
G10	Email templates (subject + text)	Notifications	Templates stored in an Email Template table
G11	What goes into the PDF export	PDF flow	Request summary + feasibility + approval + closure
G12	Recurring request rules (how often, auto-create?)	Data + flow	Frequency (Monthly/Quarterly/Yearly) and a "Create next occurrence" button; no auto-creation
G13	Exact KPI definitions in Power BI	Reports	Defined in doc 09
G14	Number of users, requests per year	Licenses, performance	~200 users, ~500 requests/year
G15	Master data files (users, departments, cost centers, locations, vendors)	Customer to provide (slide 8)	Sample CSVs in <code>sample-data/</code>

#	Missing item	Why it matters	Our ASSUMPTION in this guide
G16	Environments and licenses	Dataverse needs Power Apps Premium licenses	Dev, Test, Prod environments; premium license for every app user
G17	Who does UAT and sign-off	Slide 8 says business users do UAT	3–5 key users
G18	Branding / logo / colours	UI design	LAM logo, simple corporate colours
G19	Language, time zone, currency	Data format	English, USD, customer's local time zone
G20	Can a requester cancel their own request?	Status rules	Yes, while status is Draft, Submitted or More Info Required

5. Questions to ask your manager / customer (copy and send)

Copy this list into an email. Ask for answers before the end of Week 1.

Subject: LAM TVRMS – Clarification questions for Solution Blueprint

Hi <Manager name>,

I have gone through the LAM TV Request Management System solution approach.
To complete the Solution Blueprint, I need the following details:

A. Process

1. What does "APPG" stand for, and who are the APPG users (names / group)?
2. Please share the complete list of request statuses and who can move a request between them.
3. For "Repeat" requests – should the system copy data from an old TVID?
4. For "Recurring" requests – what frequencies are allowed, and should the system auto-create the next request?
5. What is the exact difference between 1st and 2nd Internal Alignment? Who takes part?
6. How many feasibility options (research institutions / vendors) are normally compared? Is there a scoring method?
7. Funding approval: single approver or multi-level? Any amount limits?
8. Can a request be put "On Hold"? Who can cancel a request?

B. Rules / timings

9. After how many days should reminders be sent? After how many days escalated, and to whom?
10. After how many days of no response should a request be auto-cancelled?
11. Working days only or calendar days?

C. Data and documents

12. Please share the current request template (PPT/Excel) so I can map all fields.
13. Which technical documents are mandatory at submission?
14. Please confirm the SharePoint folder structure for each TVID.
15. Please share master data: users & roles, departments, cost centers + owners, locations, research institutions/vendors.

D. Notifications & reports

16. Please share/approve the email texts for each notification.
17. What should the exported PDF contain?
18. Which KPIs are most important for the Power BI dashboard? Any targets (e.g. cycle time < 30 days)?

E. Technical

19. Which tenant/environments will we use (Dev/Test/Prod)? Who gives access?
20. Are Power Apps Premium and Power BI Pro licenses available for all users?
21. Expected number of users and requests per year?

Thanks,
<Your name>

6. Recommendation

1. **Do not wait** for all answers. Set up environments and build the data model using the assumptions in this guide (they are easy to change).
2. Hold **2–3 workshops** in Week 1 (Process, Data & Documents, Reports).
3. Write the **Solution Blueprint** in Week 2 and get it **signed off** (slide 6 says "Report submitted for final approval"). Only then start Sprint 1.
4. Any new idea after sign-off = **Change Request** (slide 10: fixed price, scope changes affect cost and schedule).

01 – Project Understanding (simple explanation)

1. The problem today

LAM's engineers need **Test Vehicles (TV)** – for example, a test wafer run, a prototype, or a lab test done at a **Research Institution** or by a vendor. Today they ask for it like this:

- They send an **email** to the APPG team.
- They attach a **PowerPoint** with the idea and an **Excel** with costs.
- Quotes from vendors come back by email.
- Funding is approved by email.
- Nobody knows the current status without asking someone.
- Documents are scattered in mailboxes and personal folders.

Result: delays, missing information, no history, no reports.

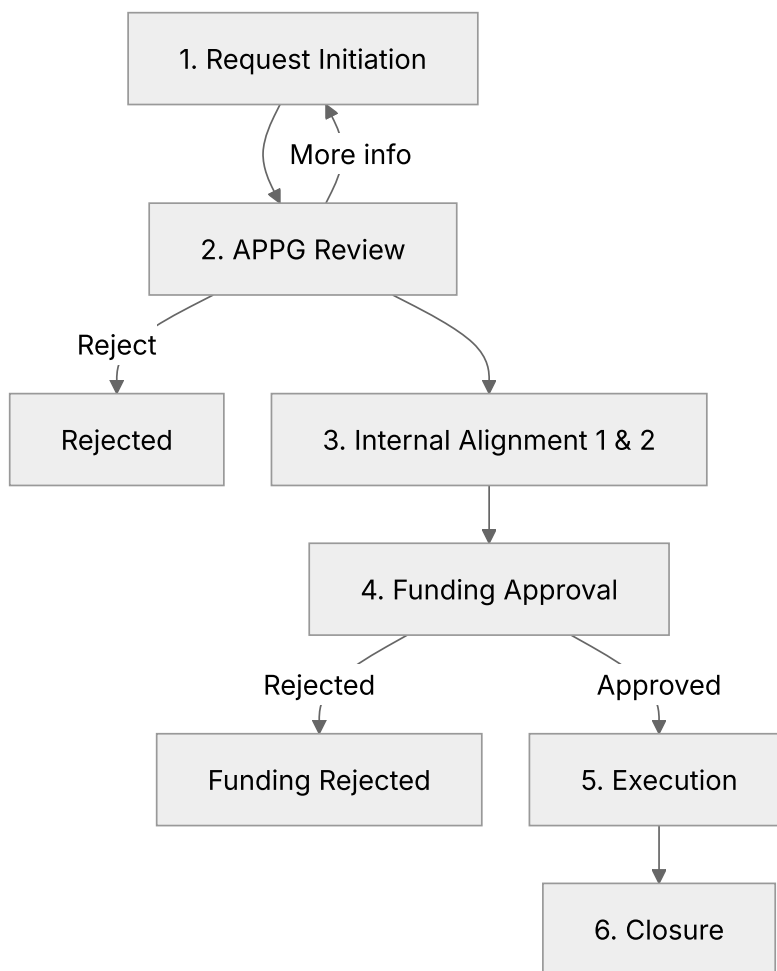
2. The solution

One application where every TV request gets a **unique ID (TVID)**, for example **TV-2026-00015**, and travels through **6 phases**. Every document is in one SharePoint folder for that TVID. Every step sends an email automatically. Managers see everything in Power BI.

3. The people (roles)

Role	Who is this?	What can they do?	Example person (sample data)
Requester	Engineer who needs a test vehicle	Create, submit, update own request, answer questions, cancel own draft	Priya Sharma (Process Engineering)
APPG User / Reviewer	Central team that reviews requests (ASSUMPTION: planning & governance team)	Review, assign owner, ask more info, reject, do feasibility, alignment, track execution, close	David Chen
Cost Center Owner (CCO)	Manager who owns the budget	Approve / reject funding	Maria Lopez (CC-4100)
Agreed Stakeholder	People who must see the request	Read-only view	Tom Baker (Program Manager)
System Administrator	IT / Power Platform admin	Master data, settings, logs, security	Admin Team

4. The 6 phases (from slide "Business Process Flow")



Phase	User journey	Key actions	Email sent
1. Request Initiation	Requester creates and submits a new TV request	Enter details & justification, upload documents, submit for APPG review, system checks mandatory fields	Acknowledgement of submission
2. APPG Review	APPG reviews and does first feasibility check	Review details, feasibility assessment, request more info or reject, submit for internal alignment	Review / More info / Rejection
3. Internal Alignment	Quotes, funding review and confirmation	1st: feasibility, quotes, estimates, clarify terms. 2nd: confirm selected option, execution path, requirements, raise funding approval request	Alignment update, Funding approval request
4. Funding Approval	Funding approved and cost center confirmed	Cost Center decision. If rejected → requester told with reason. If approved → execution starts	Funding approved / rejected
5. Execution	Work is done and tracked	Execute, track progress & activities, updates, cost & utilization tracking, exceptions & issues	Status change, reminders, escalations, exceptions
6. Closure	Request closed, results stored	Execution completed, final report uploaded to SharePoint, closure notification, audit trail	Completion, report available

5. Glossary (words you will see)

Word	Meaning
TV	Test Vehicle – the thing/test being requested
TVID	Unique request number, e.g. TV-2026-00015

Word	Meaning
APPG	The reviewing team (full form to be confirmed)
RI	Research Institution – an external lab/university/vendor that can do the test
Feasibility	"Can it be done, by whom, at what cost and time?"
CCO	Cost Center Owner – approves the money
Dataverse	Microsoft's database inside Power Platform
Canvas App	A Power App where you design every screen yourself (for end users)
Model-Driven App	A Power App built automatically from Dataverse tables (for admins)
Flow	An automation in Power Automate
Solution	A "package" in Power Platform that holds all your components so you can move them to Test/Prod
UAT	User Acceptance Testing – business users test before go-live
Hypercare	Extra support for 2 weeks after go-live
RBAC	Role-Based Access Control

6. Request types (from scope slide)

Type	Meaning	Example
New	First time request for a new test	"New plasma etch test on 300mm wafers"
Repeat	Same as a previous request; copy from old TVID	"Repeat TV-2026-00003 with new chamber material"
One-time	Done only once	"One-time thermal stress test"
Recurring	Happens again and again (monthly/quarterly)	"Quarterly contamination check at University lab"

7. What is NOT in scope (do not build these)

- Integration with any external system (e.g. SAP, ERP).
- Migrating old requests from Excel/email.
- Training business users.
- Change management.

02 – Environment Setup (do this first)

Step 1 – Check licenses (slide 8: LAM provides licenses from Day 1)

Who	License needed	Why
Every app user (requesters, APPG, CCO, stakeholders)	Power Apps Premium (per user) or Per-App	The app uses Dataverse , which is a premium connector
Every report viewer	Power BI Pro (or Premium capacity)	To view shared dashboards
Developers	Power Apps Premium + Power BI Pro (or Developer Plan for Dev)	Build and test
Flow owner (service account)	Power Automate Premium (or covered by Power Apps license in context of the app)	Flows use Dataverse + SharePoint + Outlook

Tip: Use a **service account** (e.g. `svc-tvrms@lam.com`) as owner of all flows and connections. Then emails come from one mailbox and flows don't stop if a developer leaves.

Step 2 – Create 3 environments

Go to <https://admin.powerplatform.microsoft.com> → **Environments** → **+ New**.

Name	Type	Dataverse	Used for
LAM-TVRMS-DEV	Sandbox	Yes	Developers build here
LAM-TVRMS-TEST	Sandbox	Yes	Testing & UAT
LAM-TVRMS-PROD	Production	Yes	Real users

Settings for each: Language = English, Currency = USD, Security group = an Entra ID group like `SG-TVRMS-Users` (so only these people can enter the environment).

Step 3 – Turn on auditing (NFR: all actions auditable, 7 years)

Power Platform admin center → Environment → **Settings** → **Audit and logs** → **Audit settings**:

- ☒ Start auditing
- ☒ Log access
- Retain logs: **Custom** → **2555 days** (= 7 years) (ASSUMPTION; confirm with compliance)

Step 4 – Create a Publisher and a Solution (in DEV)

1. Go to <https://make.powerapps.com> → choose `LAM-TVRMS-DEV`.
2. **Solutions** → **New solution**.
3. Click **New publisher**:
 - Display name: `LAM TVRMS`
 - Name: `lamtvrms`
 - Prefix: `tvr` → every table/column will start with `tvr_` (e.g. `tvr_tvrequest`)
4. Solution name: `TV Request Management System`, Name: `TVRMS`, Version `1.0.0.0`.

Rule: Always create everything (tables, apps, flows, roles, env variables) **inside this solution**. Never in "Default Solution".

Step 5 – Create the SharePoint site

1. Go to SharePoint → **Create site** → **Team site** (or Communication site).
2. Name: **TV Request Management** → URL e.g. `https://lam.sharepoint.com/sites/TVRMS`
3. Create a document library: **TV Requests** (this holds one folder per TVID).
4. Do one site per environment (ASSUMPTION): `TVRMS-DEV`, `TVRMS-TEST`, `TVRMS` (prod).

Details in [04-SharePoint-Document-Management](#).

Step 6 – Create Environment Variables (inside the solution)

Environment variables let the same solution point to different sites in Dev/Test/Prod.

Display name	Schema name	Type	DEV value (example)
SharePoint Site URL	<code>tvr_SPSiteUrl</code>	Text	<code>https://lam.sharepoint.com/sites/TVRMS-DEV</code>
SharePoint Library	<code>tvr_SPLibrary</code>	Text	<code>TV Requests</code>
App URL	<code>tvr_AppUrl</code>	Text	<code>https://apps.powerapps.com/play/e/.../a/...</code>
APPG Team Email	<code>tvr_APPGEmail</code>	Text	<code>appg-team@lam.com</code>
Admin Email	<code>tvr_AdminEmail</code>	Text	<code>tvrms-admin@lam.com</code>

How: Solution → **New** → **More** → **Environment variable**.

Step 7 – Create Connection References

Solution → **New** → **More** → **Connection reference**. Create one each:

- `tvr_Dataverse` (Microsoft Dataverse)
- `tvr_SharePoint` (SharePoint)
- `tvr_Outlook` (Office 365 Outlook)
- `tvr_Approvals` (Approvals)
- `tvr_Users` (Office 365 Users)
- `tvr_OneDrive` (OneDrive for Business – used for PDF conversion)

All flows must use these connection references.

Step 8 – Create Entra ID security groups (ask IT)

Group	Members (sample)	Linked to
<code>SG-TRMS-Users</code>	all users	Environment access
<code>SG-TRMS-Requesters</code>	engineers	Dataverse team → role "TVR Requester"
<code>SG-TRMS-APPG</code>	APPG team	role "TVR APPG Reviewer"
<code>SG-TRMS-CCOwners</code>	cost center owners	role "TVR Cost Center Owner"
<code>SG-TRMS-Admins</code>	IT admins	role "TVR Admin"

Step 9 – Tools on your laptop (optional but useful)

- Microsoft Edge (NFR: must work in Edge)
- **Power BI Desktop** (free, from Microsoft Store)
- **Power Platform CLI** (`pac`) for exporting solutions to source control
- **Visual Studio Code** + "Power Platform Tools" extension
- **Git** – to keep exported solution in this repository (folder `solution/` later)

Checklist

- ☐ Licenses confirmed
- ☐ DEV / TEST / PROD environments created
- ☐ Auditing on, 7-year retention
- ☐ Publisher `tvr` + solution `TVRMS` created
- ☐ SharePoint site + library created
- ☐ Environment variables + connection references created
- ☐ Security groups created

03 – Dataverse Data Model (tables, columns, sample data)

All tables are created in solution **TVRMS** with prefix **tvr_**. How to create a table: **Solution → New → Table → Table**, give the name, then add columns.

1. Big picture – how all the tables are linked (Entity Relationship)

This section explains **how the tables are linked**: 2 short lessons (1.1 and 1.2), then **2 pictures that together show all 26 links** (1.3), a **list of the 26 links** with a simple meaning and an example (1.4), and a **third picture with real data** (1.5). Each picture has its own landscape page, so the text is big.

1.1 First learn 4 simple words

Word	Simple meaning	Example in this project
Table	A list of records – like one sheet in Excel	TV Request , Department
Row (record)	One line in a table	The request TV-2026-00015
Lookup column	A column whose value is one row of another table – like picking from a dropdown	On the request, the column <i>Department</i> holds "Process Engineering"
Relationship (link)	The link that a lookup column creates between two tables	TV Request → Department

1.2 How to read the pictures

What you see	Meaning
A short bar and the letter 1 at the end of a line	The ONE side (the "parent" table)
A crow's foot (three short lines) and the letter N	The MANY side (the "child" table): zero, one or many rows
Several branches joined to one trunk	The single bar + 1 on the trunk is the ONE end of every branch – read each branch as its own sentence
The arrow → between two table names (in the text and in the cards)	"points to" : <i>TV Request → Department</i> = "a request points to one department". It always goes from the table that holds the lookup column (N) to the ONE table
The colour of a line	The colour of the table on the ONE side: blue = master data, purple = TV Request, grey = User, green = Feasibility Option (only the <i>Selected Option</i> link)
A dashed loop under the TV Request card	The self link for <i>Repeat</i> requests (empty for <i>New</i> requests)

Read every line as a sentence. Example: *Department 1 — N TV Request* means "**one** department can have **many** requests, but **each** request belongs to **only one** department".

Golden rule: the lookup column is always stored on the **N side** (the child). So the *Department* column is created on the **TV Request** table – not on the *Department* table.

Box colours: **blue** = master data · **purple** = main table (TV Request) · **green** = child tables · **grey** (dashed) = User, the people.

1.3 The three pictures – what each one shows

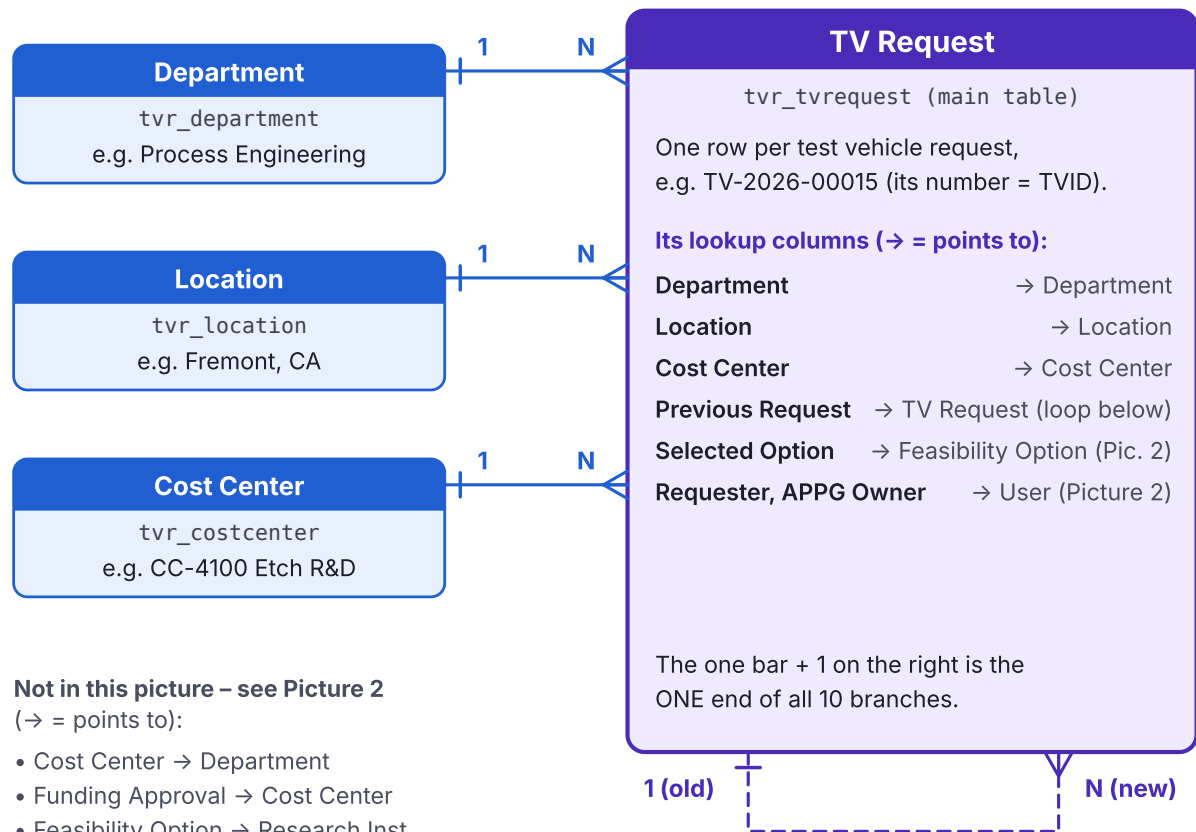
- **Picture 1 – TV Request, its 3 master-data parents, its 10 child tables and the self link (14 links).** *TV Request* is in the middle. On the left are its 3 master-data parents (Department, Location, Cost Center): each request points to one row in each. On the right are its **10 child tables**: every child table has its own branch, because every child table has a lookup column *TV Request*. The small tag on each child (Phase 3, Phase 5, Audit ...) shows when its rows are created. The dashed loop under the card is the self link for *Repeat* requests.
- **Picture 2 – the other 12 links.** Part A shows the 4 table links that are not in Picture 1: Cost Center → Department, Funding Approval → Cost Center, Feasibility Option → Research Institution, and TV Request → Feasibility Option (the lookup *Selected Option*). Part B shows the 8 "person" columns (for example *Requester* and *APPG Owner* on TV Request): all of them point to the built-in **User** table.
- **Picture 3 – real data** (in 1.5): one real request and the real rows linked to it.

Not linked to anything, so not in the pictures: **App Setting** and **Email Template** (the flows read them by key or code) and the optional **Status Transition** table (section 4).

Picture 1 – TV Request, its 3 master-data parents, its 10 child tables and the self link (14 links)

Every line is one lookup column. Bar + "1" = the ONE side. Crow's foot + "N" = the MANY side (0, 1 or many rows). → = points to.

PARENT TABLES – master data (ONE side)

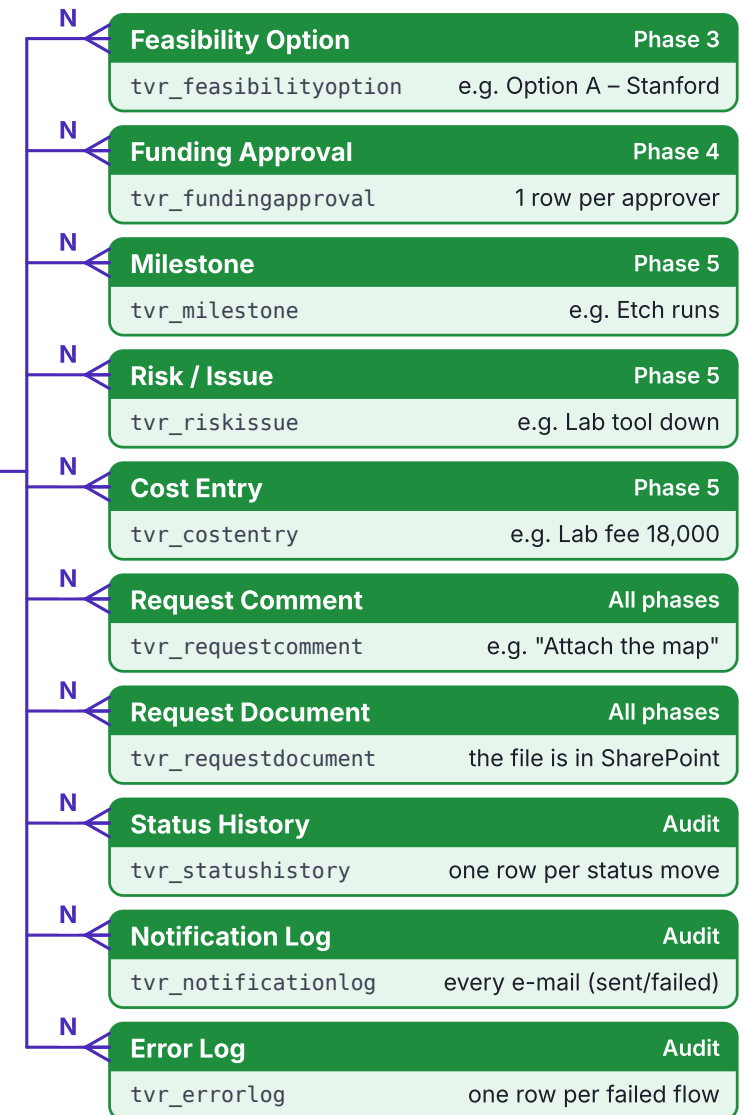


Not in this picture – see Picture 2

(→ = points to):

- Cost Center → Department
- Funding Approval → Cost Center
- Feasibility Option → Research Inst.
- TV Request → Feasibility Option (lookup "Selected Option")
- 8 person columns → User

CHILD TABLES – each has a lookup column "TV Request" (MANY side)

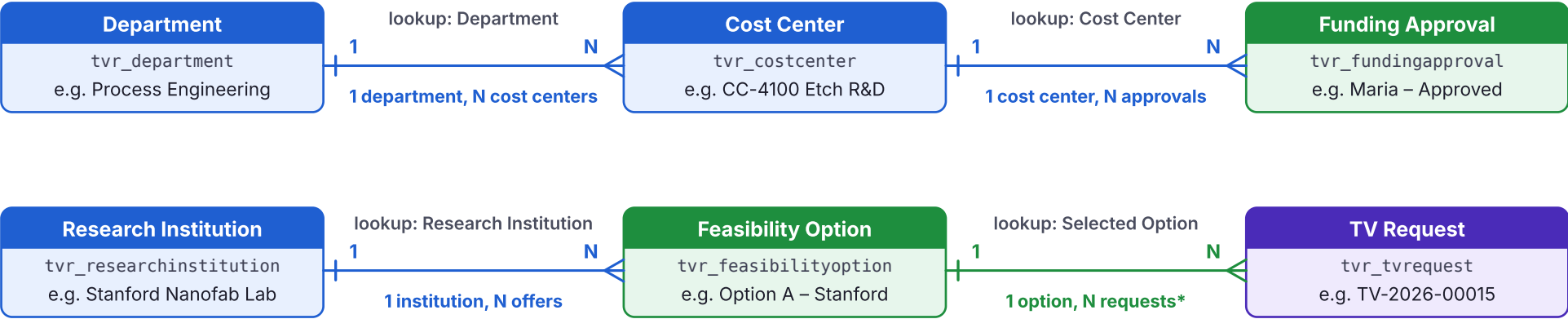


Self link "Previous Request":
a Repeat request points to the old one,
e.g. TV-2026-00016 → TV-2026-00003
(empty for New requests)

Picture 2 – The other links (12 links)

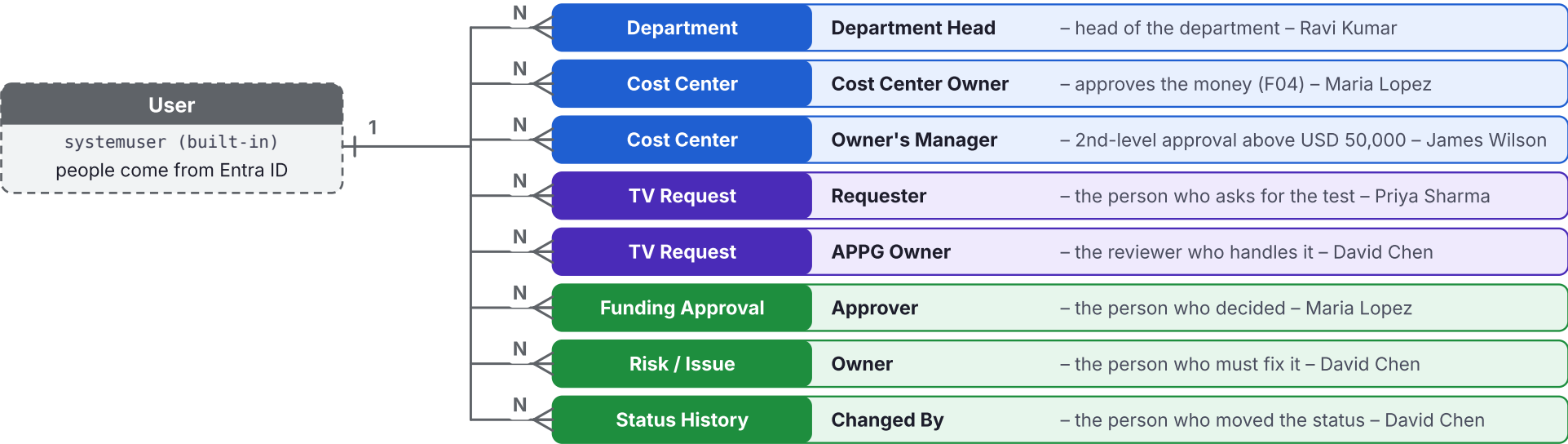
Same symbols as Picture 1. The small grey text on a line is the lookup column; it is stored on the N table.

A. THE OTHER 4 TABLE LINKS (not drawn in Picture 1)



* In practice exactly one request uses its own chosen option. The lookup is still a normal one-to-many (1 : N) link because it is stored on TV Request.

B. PERSON COLUMNS → USER (8 links) – one person can appear on many rows



1.4 Every link explained (26 links)

A. Links from master data (4 links)

#	ONE (1)	MANY (N)	Lookup column (on the N table)	What it means	Example	Picture
1	Department	TV Request	Department (<code>tvr_department</code>)	Every request belongs to one department; a department has many requests	TV-2026-00015 → Process Engineering	1
2	Location	TV Request	Location (<code>tvr_location</code>)	Where the request comes from	TV-2026-00015 → Fremont, CA	1
3	Cost Center	TV Request	Cost Center (<code>tvr_costcenter</code>)	The budget the request is charged to – its owner approves the money	TV-2026-00015 → CC-4100 Etch R&D	1
4	Department	Cost Center	Department (on Cost Center)	Each cost center belongs to one department	CC-4100 → Process Engineering	2

B. One request and its child tables (10 links). Each child table has a lookup column **TV Request** that points back to the request. The last column shows request TV-2026-00015 **at the end of the doc 13 story** (closed 5-Mar-2027).

#	ONE (1)	MANY (N)	What one child row means	TV-2026-00015 has ...	Picture
5	TV Request	Feasibility Option	One possible way to do the test (lab, cost, time)	3 rows: A Stanford 39,500 · B Fraunhofer 35,000 · C Internal Lab 22,000	1
6	TV Request	Funding Approval	One approval decision (one per approval level)	1 row: Maria Lopez – Approved – 39,500	1
7	TV Request	Milestone	One step of the work with planned and actual dates	4 rows: Wafer preparation, Etch runs, CD-SEM, Final report	1
8	TV Request	Risk / Issue	One problem or risk during execution	1 row: Lab tool down – High – Resolved	1
9	TV Request	Cost Entry	One real cost (invoice)	4 rows = 37,800, the request's <i>Actual Total Cost</i> (the sample CSV is an earlier moment: 2 rows = 23,300)	1
10	TV Request	Request Comment	One comment or question	2 rows: David asks for the wafer map, Priya answers	1
11	TV Request	Request Document	One uploaded or generated file (the file itself is in SharePoint)	7 rows in the doc 04 §5 list: spec, wafer map, quote, funding approval PDF, progress report, final report, summary PDF (more if all 3 quotes are uploaded)	1
12	TV Request	Status History	One status move (who, when, from → to)	11 rows: Draft → Submitted ... → Closed	1
13	TV Request	Notification Log	One e-mail the system tried to send (Sent or Failed)	One row per e-mail, e.g. REQ_SUBMITTED to Priya	1
14	TV Request	Error Log	One failed flow run	0 rows in this story (section 5.10 shows what a row looks like when a flow fails)	1

C. Other links between tables (4 links)

#	ONE (1)	MANY (N)	Lookup column	What it means	Example	Picture
15	Research Institution	Feasibility Option	Research Institution	Which lab or vendor made this offer. One lab can make offers for many requests	Option A → Stanford Nanofab Lab	2
16	Cost Center	Funding Approval	Cost Center	Which budget the approval is for	Maria's approval → CC-4100	2
17	TV Request (old)	TV Request (new)	Previous Request (<code>tvr_parentrequest</code>)	Self link. A <i>Repeat</i> request points to the request it repeats. Empty for <i>New</i> requests	TV-2026-00016 → TV-2026-00003	1 (dashed loop)
18	Feasibility Option	TV Request	Selected Option (<code>tvr_selectedoption</code>)	The one option chosen in Internal Alignment 2. It points to one of the request's own options (link 5)	TV-2026-00015 → Option A – Stanford	2

D. Person columns → User (8 links). In all of them **one** person can appear on **many** rows. All are in Picture 2, part B.

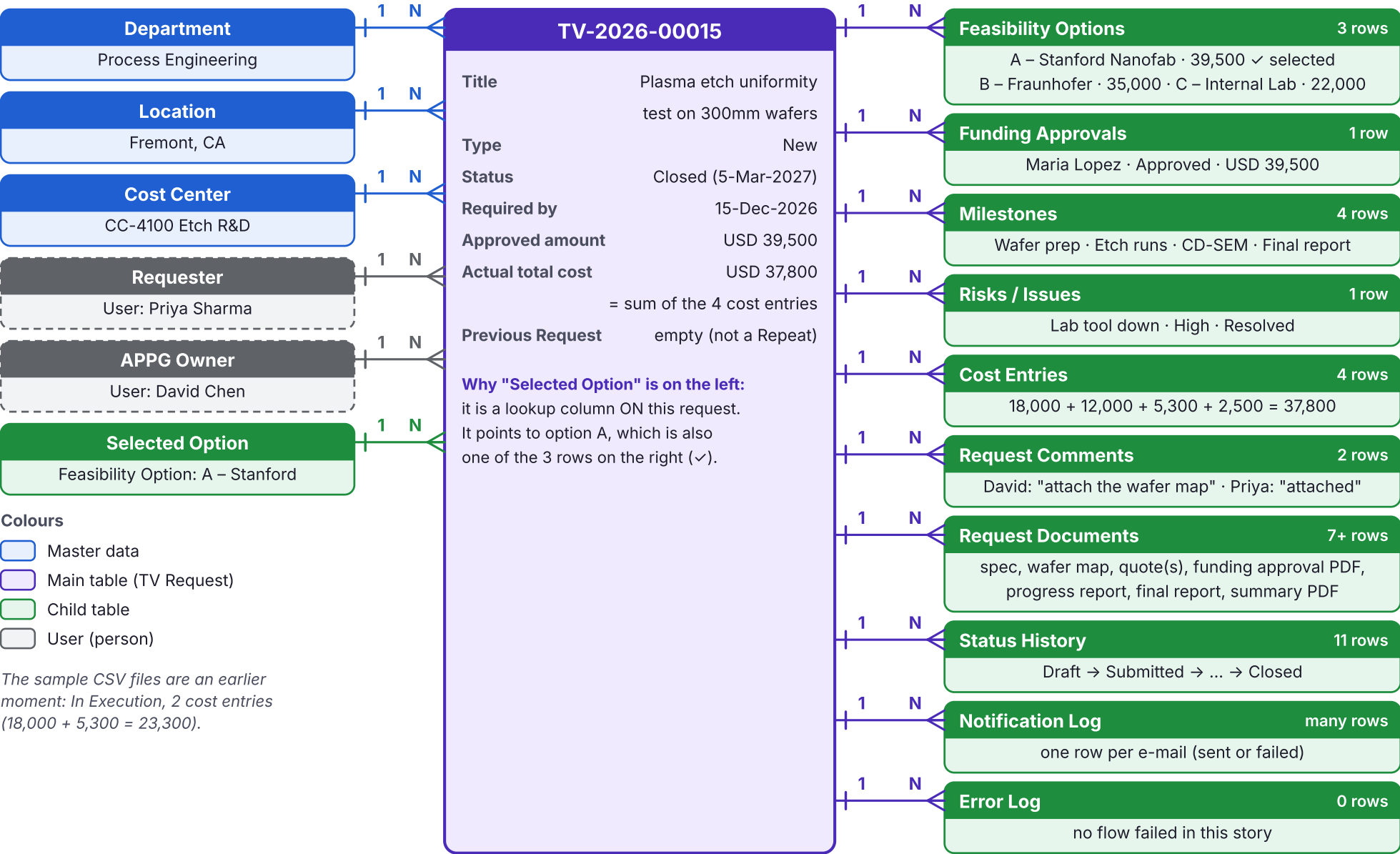
#	Table (N side)	Lookup column → User	Who is it?	Example
19	Department	Department Head	Head of the department	Ravi Kumar
20	Cost Center	Cost Center Owner	Approves the money (funding flow F04)	Maria Lopez
21	Cost Center	Owner's Manager	2nd-level approval for amounts above USD 50,000 (flow F04)	James Wilson
22	TV Request	Requester	The person who asks for the test	Priya Sharma
23	TV Request	APPG Owner	The reviewer who handles the request	David Chen
24	Funding Approval	Approver	The person who decided	Maria Lopez
25	Risk / Issue	Owner	The person who must fix the issue	David Chen
26	Status History	Changed By	The person who moved the status	David Chen

1.5 The same links with real data (Picture 3)

Picture 3 shows request **TV-2026-00015** at the end of the end-to-end use case (doc 13) and the real rows linked to it. **Left** = the one row this request points to in each parent table. **Right** = the many rows that point to this request. Note: the sample CSV files show the same request at an **earlier moment** (status *In Execution*, 2 cost entries = 23,300).

Picture 3 – Real data: request TV-2026-00015 (end of the doc 13 story, closed 5-Mar-2027)

Left: the ONE row this request points to in each parent table. Right: the MANY rows that point to this request.



1.6 How to create a link in Dataverse (step by step)

Example: link **TV Request** to **Department** (link 1).

1. Go to **make.powerapps.com** → environment **LAM-TV RMS - DEV** → **Solutions** → **TVRMS** → **Tables**.
2. Open the table on the **N side** – here **TV Request** (golden rule from 1.2).
3. Click **+ New** → **Column**. Display name: **Department**. Data type: **Lookup**. Related table: **Department**.
4. Click **Save**. Dataverse now creates the one-to-many relationship (one Department, many TV Requests) by itself.
5. Check it: open the **Department** table → **Relationships**. You see a *One-to-many* relationship to TV Request.
6. **Only for the 10 child-table links (table B):** set the behaviour from section 6 (Delete: Restrict, Assign/Share: Cascade All). For all other links (tables A, C and D) keep the **default** behaviour (no cascade). Otherwise, for example, re-assigning one Department row would also re-assign all requests of that department.

Create the lookup the same way for every row of tables A, B, C and D above (for the person columns, the related table is **User**). Common mistakes to avoid:

- Creating the lookup on the wrong table (on Department instead of TV Request). Then one department could have only one request.
- Using a text column (typing "Process Engineering") instead of a lookup. Then there is no real link, no dropdown and no report filtering.
- Creating a *many-to-many* relationship. None of the 26 links in this project needs many-to-many.

2. Master data tables (customer provides data – slide 8)

2.1 Department (**tvr_department**)

Column	Type	Required	Sample
Name (primary)	Text 100	Yes	Process Engineering
Department Code	Text 20	Yes	PE
Department Head	Lookup → User	No	Ravi Kumar
Is Active	Yes/No	Yes	Yes

2.2 Cost Center (**tvr_costcenter**)

Column	Type	Required	Sample
Name (primary)	Text 100	Yes	CC-4100 Etch R&D
Cost Center Code	Text 20	Yes	4100
Cost Center Owner	Lookup → User	Yes	Maria Lopez
Owner's Manager (for escalation / high value)	Lookup → User	No	James Wilson
Department	Lookup → Department	No	Process Engineering
Is Active	Yes/No	Yes	Yes

2.3 Location (`tvr_location`)

Column	Type	Sample
Name	Text	Fremont, CA
Country	Text	USA
Is Active	Yes/No	Yes

2.4 Research Institution / Vendor (`tvr_researchinstitution`)

Column	Type	Sample
Name	Text	Stanford Nanofab Lab
Type	Choice: University / Research Lab / Vendor / Internal Lab	University
Contact Name	Text	Dr. Alan Green
Contact Email	Email	agreen@stanford.example
Capabilities	Multiline text	Plasma etch, SEM, TEM
Is Active	Yes/No	Yes

2.5 App Setting (`tvr_appsetting`) – business can change rules without code

Name (Key)	Value	Description
ReminderAfterDays	3	Send reminder if pending action > 3 days
EscalationAfterDays	5	Escalate to manager if pending > 5 days
AutoCancelAfterDays	30	Auto-cancel if "More Info Required" > 30 days
HighValueFundingLimit	50000	Above this, 2nd-level approval
RetentionYears	7	Record retention

2.6 Email Template (`tvr_emailtemplate`)

Column	Type	Sample
Template Code (primary)	Text	REQ_SUBMITTED
Subject	Text	[TVRMS] {TVID} submitted – {Title}
Body (HTML)	Multiline	Dear {RequesterName}, Your request {TVID} has been submitted...
Is Active	Yes/No	Yes

The flow replaces `{TVID}`, `{Title}`, `{RequesterName}`, `{Status}`, `{Link}`, `{Comments}` with real values.

3. Main table – TV Request (`tvr_tvrequest`)

Turn on: **Auditing, Track changes**. Ownership: **User or team**.

#	Display name	Schema name	Type	Req.?	Sample value
1	TVID (primary)	<code>tvr_tvid</code>	Autonumber TV- {DATETIMEUTC:yyyy} -{SEQNUM:5}	auto	TV-2026-00015
2	Request Title	<code>tvr_title</code>	Text 200	Yes	Plasma etch uniformity test on 300mm wafers
3	Request Type	<code>tvr_requesttype</code>	Choice: New, Repeat, One-time, Recurring	Yes	New
4	Previous Request (for Repeat)	<code>tvr_parentrequest</code>	Lookup → TV Request	If Repeat	TV-2026-00003
5	Recurrence Frequency	<code>tvr_frequency</code>	Choice: Monthly, Quarterly, Half- yearly, Yearly	If Recurring	Quarterly
6	Requester	<code>tvr_requester</code>	Lookup → User	Yes	Priya Sharma
7	Department	<code>tvr_department</code>	Lookup → Department	Yes	Process Engineering
8	Location	<code>tvr_location</code>	Lookup → Location	Yes	Fremont, CA
9	Cost Center	<code>tvr_costcenter</code>	Lookup → Cost Center	Yes	CC-4100 Etch R&D
10	Business Justification	<code>tvr_justification</code>	Multiline 4000	Yes	Customer X reports 4% non-uniformity...
11	Technical Description	<code>tvr_technicaldescription</code>	Multiline 4000	Yes	25 wafers, 3 recipes, measure CD...
12	Expected Outcome	<code>tvr_expectedoutcome</code>	Multiline	Yes	Uniformity < 2%
13	Priority	<code>tvr_priority</code>	Choice: Low, Medium, High, Critical	Yes	High
14	Required By Date	<code>tvr_requiredbydate</code>	Date only	Yes	2026-12-15
15	Estimated Budget (USD)	<code>tvr_estimatedbudget</code>	Currency	Yes	42,000
16	Request Status (do NOT name it just "Status" – Dataverse already has a system column called Status)	<code>tvr_status</code>	Choice (see §4)	auto	Submitted
17	Current Phase	<code>tvr_phase</code>	Choice: Initiation, APPG Review, Internal Alignment, Funding Approval, Execution, Closure	auto	APPG Review
18	APPG Owner	<code>tvr_appgowner</code>	Lookup → User	at review	David Chen
19	Selected Option	<code>tvr_selectedoption</code>	Lookup → Feasibility Option	at alignment 2	Stanford Nanofab – Option A

#	Display name	Schema name	Type	Req.?	Sample value
20	Approved Amount (USD)	tvr_approvedamount	Currency	at funding	39,500
21	Expected Completion Date	tvr_expectedcompletion	Date only	execution	2027-02-28
22	Overall Progress %	tvr_progress	Whole number 0-100	execution	60
23	Actual Total Cost (USD)	tvr_actualcost	Currency (Rollup of Cost Entry)	auto	37,800
24	SharePoint Folder URL	tvr_spfolderurl	URL	auto	https://lam.sharepoint.com/sites/TVRMS/TV Requests/TV-2026-00015
25	Submitted On	tvr_submittedon	Date and time	auto	2026-10-12 10:05
26	Status Changed On	tvr_statuschangedon	Date and time	auto	used for reminders/aging
27	Closed On	tvr_closedon	Date and time	auto	2027-03-05
28	Final Outcome	tvr_finaloutcome	Multiline	at closure	Uniformity improved to 1.6%
29	Lessons Learned	tvr_lessonslearned	Multiline	at closure	Book lab 6 weeks early
30	Reuse Recommendation	tvr_reuserecommendation	Choice: Reuse as-is, Reuse with changes, Do not reuse	at closure	Reuse with changes
31	Rejection / Cancellation Reason	tvr_reason	Multiline	when rejected/cancelled	Duplicate of TV-2026-00011
32	Cycle Time (days)	tvr_cycletimedays	Formula/Calculated: DiffInDays (SubmittedOn, ClosedOn)	auto	144

4. Status list and allowed moves (ASSUMPTION – confirm in workshop)

Code	Status	Phase	Who moves it next	Allowed next status
1	Draft	Initiation	Requester	Submitted, Cancelled
2	Submitted	APPG Review	APPG	Under APPG Review
3	Under APPG Review	APPG Review	APPG	More Info Required, Rejected, Internal Alignment 1
4	More Info Required	APPG Review	Requester	Under APPG Review (resubmit), Cancelled (auto after 30 days)
5	Rejected	APPG Review	– (final)	–
6	Internal Alignment 1	Internal Alignment	APPG	Internal Alignment 2
7	Internal Alignment 2	Internal Alignment	APPG	Pending Funding Approval
8	Pending Funding Approval	Funding Approval	CCO (via flow)	Funding Approved, Funding Rejected

Code	Status	Phase	Who moves it next	Allowed next status
9	Funding Rejected	Funding Approval	– (final)	–
10	Funding Approved	Funding Approval	APPG	In Execution
11	In Execution	Execution	APPG	On Hold, Execution Completed
12	On Hold	Execution	APPG	In Execution, Cancelled
13	Execution Completed	Closure	APPG	Closed
14	Closed	Closure	– (final)	–
15	Cancelled	any	– (final)	–

Tip: put the "allowed next status" rules in **one place** (the Canvas app uses a collection or a small table `tvr_statustransition`) so the rules are easy to change.

5. Child tables

5.1 Feasibility Option (`tvr_feasibilityoption`)

Column	Type	Sample
Option Name (primary)	Text	Option A – Stanford Nanofab
TV Request	Lookup → TV Request	TV-2026-00015
Research Institution	Lookup	Stanford Nanofab Lab
Capability Score (1-5)	Whole number	5
Quoted Cost (USD)	Currency	39,500
Timeline (weeks)	Whole number	10
Constraints	Multiline	Lab available only from Nov
Recommendation	Choice: Recommended, Alternative, Not Recommended	Recommended
Is Selected	Yes/No	Yes
Quote Document URL	URL	link to SharePoint file

5.2 Funding Approval (`tvr_fundingapproval`)

Column	Type	Sample
Name	Text (auto)	FA-TV-2026-00015-1
TV Request	Lookup	TV-2026-00015
Cost Center	Lookup	CC-4100
Approver	Lookup → User	Maria Lopez
Approval Level	Whole number	1
Requested Amount	Currency	39,500
Decision	Choice: Pending, Approved, Rejected	Approved
Comments	Multiline	Approved from Q4 budget
Requested On / Decided On	Date time	2026-11-02 / 2026-11-03

This table is the "approval history" asked in scope.

5.3 Milestone (`tvr_milestone`)

Column	Type	Sample
Milestone Name	Text	Wafer preparation
TV Request	Lookup	TV-2026-00015
Sequence	Whole number	1
Planned Start / Planned End	Date	2026-11-10 / 2026-11-24
Actual Start / Actual End	Date	2026-11-12 / 2026-11-26
% Complete	Whole number	100
Status	Choice: Not Started, In Progress, Completed, Delayed	Completed

5.4 Risk / Issue (`tvr_riskissue`)

Column	Type	Sample
Title	Text	Lab tool down for maintenance
TV Request	Lookup	TV-2026-00015
Type	Choice: Risk, Issue, Exception	Issue
Severity	Choice: Low, Medium, High	High
Owner	Lookup → User	David Chen
Mitigation / Action	Multiline	Move run to 2nd lab
Status	Choice: Open, In Progress, Resolved	Resolved
Due Date	Date	2026-12-05

5.5 Cost Entry (`tvr_costentry`)

Column	Type	Sample
Description	Text	Lab usage fee – Nov
TV Request	Lookup	TV-2026-00015
Category	Choice: Lab Fee, Material, Shipping, Labor, Other	Lab Fee
Amount (USD)	Currency	18,000
Cost Date	Date	2026-11-30
Invoice Reference	Text	INV-88213

Create a **Rollup column** on TV Request: `Actual Total Cost = SUM(Cost Entry.Amount)` . **Utilization %** = Actual Total Cost / Approved Amount × 100 (calculated in Power BI or formula column).

5.6 Request Comment (`tvr_requestcomment`)

Column	Type	Sample
Comment	Multiline	Please attach the wafer map.
TV Request	Lookup	TV-2026-00015
Stage	Choice (same as Phase)	APPG Review
Comment Type	Choice: General, More Info Request, Requester Response, Decision	More Info Request
Created By / Created On	system	David Chen / 2026-10-13

5.7 Request Document (`tvr_requestdocument`) – metadata only, file lives in SharePoint

Column	Type	Sample
File Name	Text	Etch_Test_Spec_v1.pdf
TV Request	Lookup	TV-2026-00015
Document Type	Choice: Technical Specification, Supporting Document, Quote, Approval Evidence, Progress Report, Final Report	Technical Specification
SharePoint URL	URL	...
Folder	Text	02-Technical Documents
Uploaded On	Date time	2026-10-12

5.8 Status History (`tvr_statushistory`) – audit trail in readable form

Column	Type	Sample
Name	Text	TV-2026-00015: Submitted → Under APPG Review
TV Request	Lookup	TV-2026-00015
From Status / To Status	Choice	Submitted / Under APPG Review
Changed By	Lookup → User	David Chen
Changed On	Date time	2026-10-13 09:12
Remarks	Multiline	Picked up for review

5.9 Notification Log (`tvr_notificationlog`)

Column	Type	Sample
Subject	Text	[TVRMS] TV-2026-00015 submitted
TV Request	Lookup	TV-2026-00015
Template Code	Text	REQ_SUBMITTED
Sent To	Text	priya.sharma@lam.com
Sent On	Date time	2026-10-12 10:06
Result	Choice: Sent, Failed	Sent
Error Message	Multiline	(empty)

5.10 Error Log (`tvr_errorlog`) – for "Exception Remediation"

Column	Type	Sample
Flow Name	Text	F01 - On Request Submitted
TV Request	Lookup	TV-2026-00015
Error	Multiline	SharePoint: folder already exists
Run URL	URL	link to failed flow run
Status	Choice: New, Fixed, Ignored	New

6. Relationships – important settings

For all child tables → TV Request lookups, set relationship behavior:

- **Delete: Restrict** (never lose history; 7-year retention).
- **Assign/Share: Cascade All** (so when the request is shared with a CCO or stakeholder, children are shared too).

7. Business rules / validations in Dataverse (server side)

Even if the Canvas app checks fields, also add these (so data is safe from any entry point):

1. **Business Rule** on TV Request: if Request Type = Repeat → *Previous Request* is required.
2. **Business Rule**: if Request Type = Recurring → *Recurrence Frequency* is required.
3. **Business Rule**: Required By Date must be after today (show error).
4. Column *Estimated Budget* min value = 0.

8. How to load sample data

Files in `/sample-data`. Load in this order (because of lookups):

1. `departments.csv`
2. `locations.csv`
3. `research_institutions.csv`
4. `cost_centers.csv`
5. `app_settings.csv`
6. `email_templates.csv`
7. `tv_requests.csv`
8. `feasibility_options.csv`
9. `milestones.csv`
10. `cost_entries.csv`

How: open the table → **Import** → **Import data from Excel/CSV** → map columns → Import. (Users must exist in the environment first – they come from Entra ID.)

04 – SharePoint Document Management

Scope slide says: "SharePoint Document Management with an automatically created TVID workspace, predefined folder structure, versioning, permissions, and seven-year retention."

1. Structure

```

Site:      https://lam.sharepoint.com/sites/TVRMS
Library: TV Requests
├── TV-2026-00015      ← created automatically on submit (the "TVID workspace")
│   ├── 01-Request      ← request summary PDF
│   ├── 02-Technical Documents ← mandatory spec, drawings
│   ├── 03-Feasibility & Quotes ← quotes from research institutions
│   ├── 04-Funding Approval ← approval evidence PDF
│   ├── 05-Execution      ← progress reports, data
│   └── 06-Final Report & Closure ← final report, closure PDF
  
```

(ASSUMPTION: folder names – confirm with business.)

2. Library settings (do once per environment)

- Versioning:** Library → Settings → *Versioning settings* → "Create major versions" ☒, keep e.g. **50** versions. "Require check out" ☒.
- Metadata columns** on the library (helps search and retention):
 - TVID (Single line text)
 - DocumentType (Choice: Technical Specification, Supporting Document, Quote, Approval Evidence, Progress Report, Final Report)
 - Phase (Choice)
- Permissions:**
 - Stop inheriting from the site.
 - SG- TVRMS-Admins → Full Control
 - Service account svc-tvrms → Full Control (used by flows)
 - SG- TVRMS-APPG → Contribute
 - SG- TVRMS-Users → **Read** at library level (ASSUMPTION: simple model).
 - If business wants requesters to see **only their own folders**, the flow breaks inheritance on each TVID folder and grants Edit to requester + APPG (see flow F01 step 6). This is stricter but more work; decide in workshop.
- Users upload through the app**, not directly in SharePoint (keeps metadata correct).

3. Seven-year retention

Ask the **Microsoft 365 compliance admin** (Microsoft Purview):

- Purview portal → **Data lifecycle management** → **Retention labels** → **Create label**
 - Name: TVRMS-7-Years
 - Retain items for **7 years** from **when created** (or *when labelled*), then **"Start a disposition review"** (safer than auto-delete).
- Publish the label → **Auto-apply** or set as **default label** for library TV Requests (Library settings → *Apply label to items in this list or library*).

For Dataverse records: we **never delete** (Delete = Restrict), and auditing retention = 7 years (doc 02).

4. How a file goes from the app to SharePoint (simple pattern)

Canvas apps cannot write directly into SharePoint folders easily. So we use a flow:

```
Canvas App (Attachment control)
└─ Power Automate flow "F09 - Upload Document" (Power Apps V2 trigger, input = File + TVID + DocType)
    └─ Create file in /TV Requests/{TVID}/{Folder}
    └─ Update file properties (TVID, DocumentType, Phase)
    └─ Add row to Dataverse "Request Document" table with file URL
```

Details of the flow are in [08-Power-Automate-Flows](#).

5. Sample: which document goes where

Document	Uploaded by	Phase	Folder
Etch_Test_Spec_v1.pdf	Requester	Initiation	02-Technical Documents
Wafer_Map.xlsx	Requester	Initiation (after "More info")	02-Technical Documents
Stanford_Quote_Q-1182.pdf	APPG	Internal Alignment 1	03-Feasibility & Quotes
TV-2026-00015_Funding_Approval.pdf	Flow	Funding Approval	04-Funding Approval
Progress_Report_Dec.pptx	APPG	Execution	05-Execution
Final_Report.pdf	APPG	Closure	06-Final Report & Closure
TV-2026-00015_Summary.pdf	Flow	Closure	06-Final Report & Closure

05 – Security (Role-Based Access Control)

Scope: "Role-Based Access Control for Requesters, APPG users, Cost Center Owners, agreed stakeholders, and system administrators." NFR: "enforce role-based access controls to prevent unauthorized access to records, documents, and application functionality."

Security works at **3 levels**:

1. **Entra ID** – who can sign in (security group on the environment).
2. **Dataverse security roles** – which rows a person can read/write (this is the real protection).
3. **Canvas app** – which buttons/screens a person sees (only for user experience, not real security).

1. Create security roles

Go to **Power Platform admin center** → **Environment** → **Settings** → **Users + permissions** → **Security roles**. Tip: **copy** the "Basic User" role, then change it. Add each role to the solution.

Legend: ● Organization (all rows) · ● Business Unit · ● User (own rows / shared) · ✗ none

TVR Requester

Table	Create	Read	Write	Delete	Append	Append To
TV Request	●	●	●	✗	●	●
Request Comment, Request Document	●	●	●	✗	●	●
Feasibility, Funding, Milestone, Risk, Cost	✗	●	✗	✗	✗	✗
Status History	✗	●	✗	✗	✗	✗
Department, Location, Cost Center, Research Institution, Email Template, App Setting	✗	●	✗	✗	✗	●

Result: Priya sees **only her own** requests (plus ones shared with her).

TVR APPG Reviewer

Table	Create	Read	Write	Delete	Append	Append To
TV Request and all child tables	●	●	●	✗	●	●
Master tables	✗	●	✗	✗	✗	●
Notification Log, Error Log	✗	●	✗	✗	✗	✗

TVR Cost Center Owner

Table	Read	Write	Notes
TV Request	●	✗	Sees requests shared with them by the funding flow
Funding Approval	●	●	Can record decision
Child tables	●	✗	via Cascade Share
Master tables	●	✗	

TVR Stakeholder (read only)

Read ● on TV Request + children (only rows shared with them). No write.

TVR Admin

Organization level on everything + master data write + **no delete on TV Request** (retention). System Administrator role is also given to 1–2 IT people.

2. Row-level security – how CCOs and stakeholders see a request

Use an **Access Team** (no coding):

1. Enable **Access Teams** on the TV Request table (Table properties → Advanced → "Have an access team").
2. Admin center → **Access team templates** → **New**: Name `TVR Request Team`, Entity `TV Request`, rights: **Read, Append, AppendTo**.
3. When the funding flow starts, it **adds the Cost Center Owner** to this access team (Dataverse action *Perform an unbound action* → `AddUserToRecordTeam`).
4. APPG adds "agreed stakeholders" from the Request Detail screen (a button that runs the same flow).

Sample: TV-2026-00015 access team = Maria Lopez (CCO), Tom Baker (stakeholder).

3. Assign roles to users (use teams, not individuals)

Admin center → **Teams** → **New team** → Type **Microsoft Entra ID Security Group**:

Dataverse Team	Entra group	Role
TVRMS Requesters	SG-TVRMS-Requesters	TVR Requester
TVRMS APPG	SG-TVRMS-APPG	TVR APPG Reviewer
TVRMS Cost Center Owners	SG-TVRMS-CCOwners	TVR Cost Center Owner
TVRMS Admins	SG-TVRMS-Admins	TVR Admin

Now IT just adds a person to the Entra group → access works automatically.

4. Showing/hiding things in the Canvas app

In `App.OnStart` (or `App.Formulas`), find the user's roles:

```
// App.Formulas (named formulas – recalculated automatically)
CurrentUserRecord = LookUp(Users, 'Primary Email' = User().Email);
IsAPPG = !IsEmpty(Filter(LookUp(Teams, 'Team Name' = "TVRMS APPG").'Users (teammembership_association)',
    'Primary Email' = User().Email));
IsAdmin = !IsEmpty(Filter(LookUp(Teams, 'Team Name' = "TVRMS Admins").'Users (teammembership_association)',
    'Primary Email' = User().Email));
```

Simpler alternative (ASSUMPTION – fine for this size): keep a small **"App Role Member"** table (User + Role). Then: `IsAPPG = !IsBlank(LookUp('App Role Members', User.'Primary Email' = User().Email && Role = 'Role (App Role Members)'.APPG))`

Use it like: `btnApprove.Visible = IsAPPG`.

Remember: hiding a button is NOT security. Dataverse roles are the real security.

5. Security test (sample)

Test	Login as	Expected
Requester sees only own requests	Priya	Sees TV-2026-00015, not TV-2026-00016 (John's)
Requester cannot change status to "Funding Approved"	Priya	Button hidden + Dataverse field security blocks it
CCO sees only shared requests	Maria	Sees TV-2026-00015 after funding flow starts
Stakeholder cannot edit	Tom	Form is read only, save fails
Admin edits master data	Admin	Can add new Cost Center

Field security (optional, stronger): turn on *Column security* for `tvr_status`, `tvr_approvedamount` and give Write only to APPG/Admin profiles.

06 – Canvas App (for end users)

Architecture slide: Canvas App covers **Request Intake / APPG Review / Feasibility Assessment / Funding Approval / Execution Tracking / Closure**, plus **Validation & Error Handling** and **Export to PDF**.

1. Create the app

1. make.powerapps.com → LAM-TVRMS-DEV → **Solutions** → **TVRMS** → **New** → **App** → **Canvas app**.
2. Name: **TV Request Management**, Format: **Tablet** (works well in Edge on laptop).
3. **Data** pane → Add data → Dataverse tables: TV Requests, Feasibility Options, Funding Approvals, Milestones, Risk Issues, Cost Entries, Request Comments, Request Documents, Status Histories, Departments, Cost Centers, Locations, Research Institutions, App Settings, Users.
4. Settings → turn ON **"Formula-level error management"** and **"Named formulas"**.

Naming convention for controls: `scr` screen, `gal` gallery, `frm` form, `btn` button, `txt` text input, `dd` dropdown, `cmb` combo box, `dp` date picker, `lbl` label, `att` attachment. Example: `btnSubmit`, `galMyRequests`.

2. Global choices used below

Create these as **global choices** (Solution → New → Choice) so formulas are readable: `TVR Status`, `TVR Phase`, `TVR Request Type`, `TVR Priority`. In Power Fx you write them like: `'TVR Status'.Submitted`.

3. App start-up

```
// App.Formulas
gblAppColorPrimary = ColorValue("#3B2BB3");
CurrentUserRecord = LookUp(Users, 'Primary Email' = User().Email);
IsAPPG = !IsEmpty(Filter(LookUp(Teams, 'Team Name' = "TVRMS APPG").'Users (teammembership_association)',
    'Primary Email' = User().Email));
IsAdmin = !IsEmpty(Filter(LookUp(Teams, 'Team Name' = "TVRMS Admins").'Users (teammembership_association)',
    'Primary Email' = User().Email));

// App.OnStart (deep link support: open a request directly from an email link)
If(
    !IsBlank(Param("TVID")),
    Set(varSelectedRequest, LookUp('TV Requests', TVID = Param("TVID")));
    Navigate(scrRequestDetail)
)
```

Every email link looks like: `{AppUrl}?TVID=TV-2026-00015` → opens the request directly.

4. Screens (list)

#	Screen	Who uses it	Purpose
1	<code>scrHome</code>	Everyone	My requests, counters, "New Request" button
2	<code>scrNewRequest</code>	Requester	3-step wizard to create & submit
3	<code>scrRequestDetail</code>	Everyone (content depends on role)	All information about one TVID in tabs
4	<code>scrAPPGQueue</code>	APPG	Work queue of requests to review
5	<code>scrFeasibility</code>	APPG	Compare research institutions / options
6	<code>scrFunding</code>	APPG / CCO	Send for funding, see approval history

#	Screen	Who uses it	Purpose
7	scrExecution	APPG	Milestones, risks/issues, costs, progress
8	scrClosure	APPG	Final outcome, lessons learned, close request

5. Screen 1 – Home (scrHome)

Layout: header bar (logo, title, user name), 4 counter tiles, a filter row, a gallery.

Counter tiles (labels):

```
// lblMyOpen.Text
CountRows(Filter('TV Requests', Requester.'Primary Email' = User().Email,
    !('Request Status' in ['TVR Status'.Closed, 'TVR Status'.Cancelled, 'TVR Status'.Rejected, 'TVR
        Status'.Funding Rejected'])))

// lblNeedsMyAction.Text (requests waiting for my answer)
CountRows(Filter('TV Requests', Requester.'Primary Email' = User().Email,
    'Request Status' = 'TVR Status'.More Info Required'))
```

Gallery galMyRequests.Items:

```
SortByColumns(
    Filter(
        'TV Requests',
        (IsAPPG || Requester.'Primary Email' = User().Email),
        IsBlank(txtSearch.Text) || StartsWith(TVID, txtSearch.Text) || StartsWith('Request Title',
            txtSearch.Text),
        IsBlank(ddStatusFilter.Selected.Value) || 'Request Status' = ddStatusFilter.Selected.Value
    ),
    "createdon", SortOrder.Descending
)
```

(Dataverse security already limits rows; the filter just makes the list cleaner.)

Inside the gallery show: TVID, Title, Status (coloured pill), Required By, Phase. Status colour:

```
// lblStatusPill.Fill
Switch(ThisItem.'Request Status',
    'TVR Status'.More Info Required, ColorValue("#F4B400"),
    'TVR Status'.Rejected, ColorValue("#D93025"),
    'TVR Status'.Funding Rejected, ColorValue("#D93025"),
    'TVR Status'.Closed, ColorValue("#188038"),
    ColorValue("#1A73E8"))
```

galMyRequests.OnSelect: Set(varSelectedRequest, ThisItem); Navigate(scrRequestDetail)

btnNewRequest.OnSelect: NewForm(frmRequest); Set(varStep, 1); Navigate(scrNewRequest)

6. Screen 2 – New Request wizard (`scrNewRequest`)

Use **one Edit Form** `frmRequest` (DataSource = 'TV Requests') and show fields step by step using `varStep`.

Step	Fields	Sample input
1 – Basic	Title*, Request Type*, Previous Request (if Repeat), Frequency (if Recurring), Department*, Location*, Cost Center*, Priority*, Required By*, Estimated Budget*	"Plasma etch uniformity test", New, –, –, Process Engineering, Fremont CA, CC-4100, High, 15-Dec-2026, 42000
2 – Details	Business Justification*, Technical Description*, Expected Outcome*	text
3 – Documents & Submit	Attachment control (technical docs*), other docs, Review summary, Save as Draft / Submit	Etch_Test_Spec_v1.pdf

Show/hide data cards:

```
// DataCard "Previous Request".Visible
ddRequestType.Selected.Value = 'TVR Request Type'.Repeat
// DataCard "Recurrence Frequency".Visible
ddRequestType.Selected.Value = 'TVR Request Type'.Recurring
```

Repeat request – copy data from old TVID (Previous Request combo `cmbPrevious.OnChange`):

```
If(!IsBlank(cmbPrevious.Selected),
    Set(varCopyFrom, cmbPrevious.Selected);
    Notify("Details copied from " & varCopyFrom.TVID & ". Please review.", NotificationType.Information)
)
// then each text input Default = If(!IsBlank(varCopyFrom), varCopyFrom.'Business Justification',
Parent.Default)
```

Attachment control: Dataverse forms don't give a free attachment control, so add a SharePoint/Dataverse form once, copy its Attachments card's control, paste it into `scrNewRequest`, rename it `attTechDocs`. Set `MaxAttachments = 10`, `MaxAttachmentSize = 25` (MB).

Validation + Submit (`btnSubmit.OnSelect`)

```
// 1. Validation (mandatory fields + technical document)
If(
    !frmRequest.Valid,
        Notify("Please fill all fields marked with *", NotificationType.Error),
    CountRows(attTechDocs.Attachments) = 0,
        Notify("Please attach at least one technical specification document.", NotificationType.Error),
    dpRequiredBy.SelectedDate <= Today(),
        Notify("'Required By' date must be a future date.", NotificationType.Error),
    ddRequestType.Selected.Value = 'TVR Request Type'.Repeat && IsBlank(cmbPrevious.Selected),
        Notify("Please select the previous request for a Repeat request.", NotificationType.Error),

    // 2. All good -> save the request as Draft first (gets a TVID)
    Set(varNew,
        Patch('TV Requests',
            If(IsBlank(varDraft), Defaults('TV Requests'), varDraft),
            frmRequest.Updates,
            { Requester: CurrentUserRecord, 'Request Status': 'TVR Status'.Draft, 'Current Phase': 'TVR Phase'.Initiation }
        )
    );
    IfError(varNew, Notify("Could not save: " & FirstError.Message, NotificationType.Error); Exit());

    // 3. Upload each file to SharePoint through flow F09
    ForAll(attTechDocs.Attachments As f,
        'F09-UploadDocument'.Run(varNew.TVID, "Technical Specification", "02-Technical Documents",
            { file: { name: f.Name, contentBytes: f.Value } })
    );
```

```
// 4. Change status to Submitted (this triggers flow F01)
Patch('TV Requests', varNew, { 'Request Status': 'TVR Status'.Submitted, 'Current Phase': 'TVR
Phase'. 'APPG Review', 'Submitted On': Now(), 'Status Changed On': Now() });
Patch('Status Histories', Defaults('Status Histories'),
  { Name: varNew.TVID & ": Draft → Submitted", 'TV Request': varNew,
    'From Status': 'TVR Status'.Draft, 'To Status': 'TVR Status'.Submitted,
    'Changed On': Now(), Remarks: "Submitted by requester" });

Notify("Request " & varNew.TVID & " submitted successfully.", NotificationType.Success);
Reset(attTechDocs); Set(varDraft, Blank());
Navigate(scrHome)
)
```

`btnSaveDraft.OnSelect` = same Patch with `'Request Status': 'TVR Status'.Draft` and **no** validation except Title. Store result in `varDraft` so the next save updates the same record.

7. Reusable "change status" pattern

Many screens change status. Use one hidden button `btnDoStatusChange` on the detail screen:

```
// btnDoStatusChange.OnSelect (inputs: varNewStatus, varRemarks)
With({ oldStatus: varSelectedRequest.'Request Status' },
  Set(varSelectedRequest,
    Patch('TV Requests', varSelectedRequest,
      { 'Request Status': varNewStatus, 'Status Changed On': Now(),
        'Current Phase': Switch(varNewStatus,
          'TVR Status'. 'Internal Alignment 1', 'TVR Phase'. 'Internal Alignment',
          'TVR Status'. 'Internal Alignment 2', 'TVR Phase'. 'Internal Alignment',
          'TVR Status'. 'Pending Funding Approval', 'TVR Phase'. 'Funding Approval',
          'TVR Status'. 'Funding Approved', 'TVR Phase'. 'Funding Approval',
          'TVR Status'. 'In Execution', 'TVR Phase'. Execution,
          'TVR Status'. 'On Hold', 'TVR Phase'. Execution,
          'TVR Status'. 'Execution Completed', 'TVR Phase'. Closure,
          'TVR Status'. Closed, 'TVR Phase'. Closure,
          varSelectedRequest.'Current Phase') }) // other statuses keep the same phase
    );
  Patch('Status Histories', Defaults('Status Histories'),
    { Name: varSelectedRequest.TVID & ": " & Text(oldStatus) & " → " & Text(varNewStatus),
      'TV Request': varSelectedRequest,
      'From Status': oldStatus, 'To Status': varNewStatus,
      'Changed On': Now(), Remarks: varRemarks });
  If(!IsBlank(varRemarks),
    Patch('Request Comments', Defaults('Request Comments'),
      { Comment: varRemarks, 'TV Request': varSelectedRequest, Stage: varSelectedRequest.'Current
        Phase' }));
)
```

Any button then does:

```
Set(varNewStatus, 'TVR Status'. 'More Info Required');
Set(varRemarks, txtReviewComment.Text);
Select(btnDoStatusChange)
```

Flow **F02** sees the status change and sends the right email.

8. Screen 3 – Request Detail (`scrRequestDetail`)

Top: TVID, Title, Status pill, Phase **progress bar** (6 chevrons; current phase highlighted). Tabs (use a horizontal gallery of tab names + `varTab`):

Tab	Content	Edit allowed for
Summary	Display form of the request	Requester (only in Draft / More Info Required)
Comments	Gallery of Request Comments + add box	Everyone with access
Documents	Gallery of Request Documents (click opens SharePoint link) + upload	Requester, APPG
Feasibility	Options gallery (read-only for requester)	APPG
Funding	Funding Approvals history	APPG (send), CCO (read)
Execution	Milestones, Risks/Issues, Costs	APPG
Closure	Final outcome, lessons learned, reuse	APPG
History	Status Histories, newest first	read only

Action buttons (visible by role and status):

Button	Visible when	Sets status to
Start Review	<code>IsAPPG && 'Request Status' = Submitted</code>	Under APPG Review (and APPG Owner = me)
Request More Info	<code>IsAPPG && 'Request Status' = Under APPG Review</code>	More Info Required (comment mandatory)
Reject	<code>IsAPPG && 'Request Status' in [Under APPG Review, Internal Alignment 1]</code>	Rejected (reason mandatory)
Resubmit	<code>Requester = me && 'Request Status' = More Info Required</code>	Under APPG Review
Move to Internal Alignment	<code>IsAPPG && 'Request Status' = Under APPG Review</code>	Internal Alignment 1
Confirm Option (Alignment 2)	<code>IsAPPG && 'Request Status' = Internal Alignment 1 && an option is selected</code>	Internal Alignment 2
Send for Funding Approval	<code>IsAPPG && 'Request Status' = Internal Alignment 2</code>	Pending Funding Approval (flow F04 starts)
Start Execution	<code>IsAPPG && 'Request Status' = Funding Approved</code>	In Execution
Put On Hold / Resume	<code>IsAPPG</code>	On Hold / In Execution
Complete Execution	<code>IsAPPG && 'Request Status' = In Execution</code>	Execution Completed
Close Request	<code>IsAPPG && 'Request Status' = Execution Completed && Final Report uploaded</code>	Closed (flow F07)
Cancel	<code>(Requester = me && 'Request Status' in [Draft, Submitted, More Info Required]) IsAPPG</code>	Cancelled
Export to PDF	everyone	calls flow F06

Example "Reject" with mandatory reason:

```
// btnReject.OnSelect
If(IsBlank(txtReviewComment.Text),
    Notify("Please enter the rejection reason.", NotificationType.Error),
    Patch('TV Requests', varSelectedRequest, { 'Rejection / Cancellation Reason': txtReviewComment.Text });
    Set(varNewStatus, 'TVR Status'.Rejected);
    Set(varRemarks, "Rejected: " & txtReviewComment.Text);
    Select(btnDoStatusChange)
)
```

Requester can see "Required By" in red if late:

```
lblRequiredBy.Color = If(varSelectedRequest.'Required By Date' < Today() && varSelectedRequest.'Request
Status' <> 'TVR Status'.Closed, Color.Red, Color.Black)
```

9. Screen 4 – APPG Queue (scrAPPGQueue)

```
// galQueue.Items
SortByColumns(
    Filter('TV Requests',
        'Request Status' in ['TVR Status'.Submitted, 'TVR Status'.Under APPG Review, 'TVR
        Status'.Internal Alignment 1',
        'TVR Status'.Internal Alignment 2', 'TVR Status'.Funding Approved, 'TVR Status'.In
        Execution',
        'TVR Status'.Execution Completed'],
        chkMyItems.Value = false || 'APPG Owner'.Primary Email = User().Email,
        "tvr_statuschangedon", SortOrder.Ascending) // oldest waiting first
```

Show "Days waiting": `RoundDown(DateDiff(ThisItem.'Status Changed On', Now(), TimeUnit.Hours) / 24, 0)`.

Assign owner: combo box `cmbAPPGOwner` (Items = Users filtered by APPG team) → `Patch('TV Requests', varSelectedRequest, { 'APPG Owner': cmbAPPGOwner.Selected })`.

10. Screen 5 – Feasibility (scrFeasibility)

Gallery of options for the request + "Add Option" form.

```
// galOptions.Items
Filter('Feasibility Options', 'TV Request'.TV Request = varSelectedRequest.TV Request)

// "Select this option" button inside gallery
ForAll(Filter('Feasibility Options', 'TV Request'.TV Request = varSelectedRequest.TV Request) As o,
    Patch('Feasibility Options', o, { 'Is Selected': false }));
Patch('Feasibility Options', ThisItem, { 'Is Selected': true, Recommendation: 'Recommendation (Feasibility
Options)'.Recommended });
Set(varSelectedRequest, Patch('TV Requests', varSelectedRequest, { 'Selected Option': ThisItem, 'Estimated
Budget': ThisItem.'Quoted Cost' }));
```

(Here `'TV Request'` inside `'TV Request'.TV Request'` is the GUID primary key column of the table.)

Sample comparison shown to APPG:

Option	Institution	Capability	Cost	Weeks	Constraint	Recommendation
A	Stanford Nanofab Lab	5	39,500	10	Lab free from Nov	✓ Recommended
B	Fraunhofer IPMS	4	35,000	16	Shipping to Germany	Alternative
C	Internal Lab Fremont	3	22,000	20	Tool booked till Jan	Not recommended

A small column chart (Cost vs Weeks) helps decision. Quotes are uploaded with F09 into **03-Feasibility & Quotes**.

11. Screen 6 – Funding (**scrFunding**)

- Shows Cost Center, its Owner (from master), Selected option cost, approval history gallery.
- Send for Funding Approval** button: sets status Pending Funding Approval → flow **F04** creates the approval for the CCO (in Outlook + Teams + Approvals app).
- CCO can also open the request from the email link and see everything read-only.

```
// lblCCOwner.Text
varSelectedRequest.'Cost Center'.'Cost Center Owner'.'Full Name'
// galFundingHistory.Items
SortByColumns(Filter('Funding Approvals', 'TV Request'.'TV Request' = varSelectedRequest.'TV Request'),
    "createdon", SortOrder.Descending)
```

12. Screen 7 – Execution (**scrExecution**)

Three sections:

- Milestones** – editable gallery (Planned/Actual dates, % complete). Overall progress:

```
// when a milestone is saved:
Patch('TV Requests', varSelectedRequest,
    { 'Overall Progress %': Round(Average(Filter('Milestones', 'TV Request'.'TV Request' =
        varSelectedRequest.'TV Request'), '% Complete'), 0) })
```

- Risks / Issues / Exceptions** – add & update (severity High → flow F02 sends exception email).
- Costs** – add cost entries; show **Actual Total Cost** vs **Approved Amount** and utilization %:

```
lblUtil.Text = Text(varSelectedRequest.'Actual Total Cost' / varSelectedRequest.'Approved Amount', "0%")
```

Show it in red if > 100%. Plus "Expected Completion Date" date picker and "Add update / comment" box.

13. Screen 8 – Closure (**scrClosure**)

Fields: Final Outcome*, Lessons Learned*, Reuse Recommendation*, Final Report upload* (goes to **06-Final Report & Closure**), Final cost (read only, rollout).

```
// btnCloseRequest.OnSelect
If(
    IsBlank(txtOutcome.Text) || IsBlank(txtLessons.Text) || IsBlank(ddReuse.Selected),
        Notify("Outcome, lessons learned and reuse recommendation are mandatory.", NotificationType.Error),
        IsEmpty(Filter('Request Documents', 'TV Request'.'TV Request' = varSelectedRequest.'TV Request',
            'Document Type' = 'Document Type (Request Documents)'.'Final Report')),
            Notify("Please upload the final report first.", NotificationType.Error),
            Patch('TV Requests', varSelectedRequest,
                { 'Final Outcome': txtOutcome.Text, 'Lessons Learned': txtLessons.Text,
                    'Reuse Recommendation': ddReuse.Selected.Value, 'Closed On': Now() });
                Set(varNewStatus, 'TVR Status'.Closed); Set(varRemarks, "Request closed");
                Select(btnDoStatusChange)
            )
```

14. Export to PDF

`btnExportPdf.OnSelect`:

```
Set(varPdf, 'F06-GeneratePDF'.Run(varSelectedRequest.TVID));  
Launch(varPdf.fileurl)
```

Flow F06 builds the PDF, saves it in SharePoint and returns the link.

15. Error handling rules (Validation & Error Handling)

- Every `Patch` wrapped with `IfError(..., Notify(FirstError.Message, NotificationType.Error))`.
- Disable buttons while saving: `btnSubmit.DisplayMode = If(varSaving, DisplayMode.Disabled, DisplayMode.Edit)`.
- Show a spinner when `varSaving = true`.
- Use `App.OnError` to log unexpected errors:

```
// App.OnError  
Notify("Something went wrong. Please try again or contact support.", NotificationType.Error)
```

16. Share the app

- Solution → app → **Share** with the Entra groups (`SG-TVRMS-Users`) – **not** individuals.
- Copy the app play URL into the environment variable `tvr_AppUrl` (used in email links).

07 – Model-Driven App (for Admin users)

Architecture slide: Model Driven App (Admin users) → **Master Data management** and **View Notification Logs**.

1. Create the app

1. Solution TVRMS → **New → App → Model-driven app**.
2. Name: **TVRMS Administration**.
3. Add a **navigation (sitemap)** with these areas and groups:

Area	Group	Pages (tables)
Requests	Requests	TV Requests (all views), Feasibility Options, Funding Approvals
Requests	Execution	Milestones, Risk Issues, Cost Entries
Master Data	Organisation	Departments, Cost Centers, Locations
Master Data	Partners	Research Institutions
Configuration	Settings	App Settings, Email Templates, Status Transitions (optional)
Monitoring	Logs	Notification Logs, Error Logs, Status Histories
Monitoring	Audit	Audit Summary (link to Dataverse audit history)

4. **Share** the app only with the **TVRMS Admins** team / role **TVR Admin**.

2. Views to create (sample)

Table	View name	Filter	Columns
TV Requests	Active Requests	Status not in (Closed, Cancelled, Rejected, Funding Rejected)	TVID, Title, Requester, Status, Phase, Status Changed On, Required By
TV Requests	Overdue Requests	Required By < Today AND Status not Closed	TVID, Title, Requester, Required By, APPG Owner
Cost Centers	Active Cost Centers	Is Active = Yes	Code, Name, Owner, Owner's Manager
Notification Logs	Failed Notifications	Result = Failed	Subject, Sent To, Sent On, Error
Error Logs	Open Errors	Status = New	Flow Name, TVID, Error, Run URL

3. Forms

- **Cost Center main form:** Code, Name, Owner, Owner's Manager, Department, Is Active. Business rule: Owner is required.
- **App Setting form:** Key (read-only after create), Value, Description.
- **Email Template form:** Code, Subject, Body (use the rich text control), Is Active.
- **TV Request main form** (admin view): all fields in tabs + sub-grids for each child table + **Audit History** tab (built in).

4. Typical admin tasks (use cases)

Use case	Steps	Sample
New cost center created by Finance	Master Data → Cost Centers → New → fill → Save	CC-4300 "Deposition R&D", Owner = Kevin Brown
Cost center owner changes	Open CC-4100 → change Owner → Save. New funding requests go to new owner	Maria Lopez → Anna White
Change reminder days	Configuration → App Settings → ReminderAfterDays → 2 → Save	No code change needed
Email failed	Monitoring → Failed Notifications → check error → fix email address → re-run from flow history	"Mailbox not found"
See who changed a request	Open request → Related → Audit History	Status changed by David Chen on 13-Oct
Deactivate a vendor	Research Institution → Is Active = No	"Old Lab Inc."

Do not delete master data – set **Is Active = No**. Dropdowns in the Canvas app filter on `Is Active = true`, for example: `cmbCostCenter.Items = Filter('Cost Centers', 'Is Active' = true)`.

08 – Power Automate Flows (all automation)

Architecture slide lists these automation activities. Here is which flow does what:

Activity from slide	Flow
SharePoint workspace creation	F01
Submission notification	F01
Missing information notification	F02
Funding approval routing	F04
Status change notifications	F02
Reminders & escalations (+ automatic cancellation)	F05
Exception remediation	F08 pattern + Error Log
Completion notifications	F07
Generate PDF and store	F06
Audit trail capture	Dataverse auditing + Status History rows (app & flows) + Notification Log
Document upload from app	F09

General rules for every flow

1. Create it **inside the solution** (Solution → New → Automation → Cloud flow).
2. Use **connection references** and **environment variables** (doc 02).
3. Owner = service account `svc-tvrms`.
4. Name format: `TVRMS - F01 - On Request Submitted`.
5. Use **Try / Catch scopes** (see F08).
6. Choice values in Dataverse are numbers. Write them down once, for example:

Status	Value (example – check yours in the choice editor)
Draft	100000000
Submitted	100000001
Under APPG Review	100000002
More Info Required	100000003
Rejected	100000004
Internal Alignment 1	100000005
Internal Alignment 2	100000006
Pending Funding Approval	100000007
Funding Rejected	100000008
Funding Approved	100000009
In Execution	100000010
On Hold	100000011
Execution Completed	100000012
Closed	100000013
Cancelled	100000014

Common building block: "Send templated email + log it"

Used inside many flows (copy the steps, or make it a child flow **F00 - Send Notification** with inputs: TemplateCode, TVRequestId, ToEmails, ExtraComments).

1. **List rows** (Dataverse) – table *Email Templates*, Filter rows: `tvr_templatecode eq 'REQ_SUBMITTED'` and `tvr_isactive eq true`, Row count 1.

2. **Compose – Subject:**

```
replace(replace(first(outputs('List_template'))['body/value'],
  '{TVID}', triggerOutputs()['body/tvr_tvid']),
  '{Title}', triggerOutputs()['body/tvr_title'])
```

3. **Compose – Body:** same `replace(...)` for `{TVID}`, `{Title}`, `{RequesterName}`, `{Status}`, `{Comments}`, and `{Link}` = `concat(parameters('App URL (tvr_AppUrl)'), '?TVID=', triggerOutputs()['body/tvr_tvid'])`.

4. **Send an email (V2)** (Office 365 Outlook) – To, Subject, Body (HTML). Tip: use **"Send an email from a shared mailbox (V2)"** with `tvrms-noreply@lam.com`.

5. **Add a new row – Notification Logs:** Subject, Template Code, Sent To, Sent On = `utcNow()`, Result = Sent, TV Request = `tvr_tvrequests(<id>)`.

6. If step 4 fails (Configure run after → *has failed*): add Notification Log with Result = Failed + error text.

Sample email (REQ_SUBMITTED)

Subject: [TVRMS] TV-2026-00015 submitted – Plasma etch uniformity test on 300mm wafers

Dear Priya Sharma,

Your Test Vehicle request TV-2026-00015 has been submitted successfully and is now with the APPG team for review.

Request type : New
Required by : 15-Dec-2026
Status : Submitted

Open request: <https://apps.powerapps.com/play/e/.../a/...?TVID=TV-2026-00015>

Regards,
TV Request Management System (automatic email – please do not reply)

F01 – On Request Submitted

Purpose: create SharePoint workspace, send acknowledgement, inform APPG.

Trigger: Dataverse – *When a row is added, modified or deleted*

- Change type: **Modified**
- Table: TV Requests
- Scope: Organization
- Select columns: `tvr_status`
- Filter rows: `tvr_status eq 100000001` (Submitted)

Steps:

1. **Scope – Try**
2. **Initialize variable** `FolderList` (Array): `["01-Request", "02-Technical Documents", "03-Feasibility & Quotes", "04-Funding Approval", "05-Execution", "06-Final Report & Closure"]`
3. **Apply to each** `FolderList` → **SharePoint Create new folder**
 - Site: env var `tvr_SPSiteUrl`, List: `tvr_SPLibrary`
 - Folder path: `@{triggerOutputs()?['body/tvr_tvid']}/@{items('Apply_to_each')}`
4. **Update a row** – TV Requests: `SharePoint Folder URL` = `concat(parameters('SharePoint Site URL (tvr_SPSiteUrl)'), '/', parameters('SharePoint Library (tvr_SPLibrary)'), '/', triggerOutputs()?['body/tvr_tvid'])`
5. **Get a row by ID** – Users: `requester ( _tvr_requester_value )` → email + full name.
6. (Optional, stricter security) **SharePoint Send an HTTP request** to break inheritance on the TVID folder and grant Edit to requester (see doc 04 §2).
7. Send templated email **REQ_SUBMITTED** → requester.
8. Send templated email **NEW_REQUEST_APPG** → env var `tvr_APPGEmail`.
9. **Scope – Catch** (run after Try *has failed / timed out*) → F08 pattern.

Test with sample: submit TV-2026-00015 → folder `TV Requests/TV-2026-00015/...` with 6 sub-folders exists, Priya gets email, APPG mailbox gets email, 2 Notification Log rows.

F02 – Status Change Notifier

Purpose: one flow that sends the right email for every status change.

Trigger: Dataverse *When a row is modified* – TV Requests – Select columns `tvr_status` (no filter).

Steps:

1. Get requester, APPG owner, and cost center owner emails (Get a row by ID ×3; use *Configure run after* to continue if a lookup is empty).
2. Get the latest comment (List rows *Request Comments*, filter on this request, order by `createdon desc`, top 1) → used as `{Comments}`.
3. **Switch** on `triggerOutputs()?['body/tvr_status']`:

Case (status)	Template	To	CC
Under APPG Review	REVIEW_STARTED	Requester	–
More Info Required	MORE_INFO	Requester	APPG owner
Rejected	REJECTED	Requester	APPG team
Internal Alignment 1 / 2	ALIGNMENT_UPDATE	Requester, stakeholders	APPG owner
Funding Approved	FUNDING_APPROVED	Requester, APPG owner	CCO
Funding Rejected	FUNDING_REJECTED	Requester, APPG owner	CCO
In Execution / On Hold	STATUS_UPDATE	Requester, stakeholders	–
Execution Completed	STATUS_UPDATE	Requester	–

Case (status)	Template	To	CC
Cancelled	CANCELLED	Requester	APPG owner
Submitted, Pending Funding Approval, Closed	<i>(do nothing – F01/F04/F07 handle these)</i>		

4. Send templated email + log (common block).

Exception notification: a second small trigger flow **F02b** on *Risk Issues* "When a row is added" with filter `tvr_severity eq <High>` → email APPG owner + requester with template EXCEPTION_RAISED.

F04 – Funding Approval Routing

Trigger: TV Requests modified, Select columns `tvr_status`, Filter `tvr_status eq 100000007` (Pending Funding Approval).

Steps:

1. **Get a row by ID** – Cost Centers (`_tvr_costcenter_value`) → Owner, Owner's Manager.
2. **Get a row by ID** – Users → CCO email, Manager email.
3. **Get a row by ID** – Feasibility Options (selected option) → quoted cost, institution.
4. **List rows** – App Settings, filter `tvr_name eq 'HighValueFundingLimit'` → limit (e.g. 50000).
5. **Perform an unbound action** `AddUserToRecordTeam` → add CCO to the request's access team (so CCO can open the record) – doc 05 §2.
6. **Add a new row** – Funding Approvals: Request, Cost Center, Approver = CCO, Level 1, Requested Amount = quoted cost, Decision = Pending, Requested On = `utcNow()`.
7. **Start and wait for an approval** (Approvals connector)
 - Type: **Approve/Reject – First to respond**
 - Title: `Funding approval: @{TVID} – USD @{formatNumber(QuotedCost,'N0')}`
 - Assigned to: CCO email
 - Details (markdown):

```

**Request:** TV-2026-00015 – Plasma etch uniformity test
**Requester:** Priya Sharma (Process Engineering)
**Cost center:** CC-4100 Etch R&D
**Selected option:** Stanford Nanofab Lab – USD 39,500 – 10 weeks
**Justification:** Customer X reports 4% non-uniformity...

```

- Item link: app deep link `?TVID=...`

8. **Update a row** – Funding Approval: Decision = outcome, Comments = `first(body('Start_and_wait_for_an_approval')?['responses'])?['comments']`, Decided On = `utcNow()`.
9. **Condition:** Outcome = "Approve" **and** amount > limit?
 - Yes → repeat steps 6–8 with **Level 2**, Approver = Owner's Manager.
10. **Condition:** final outcome = Approve?
 - **Yes:** Update TV Request → Status = Funding Approved, Approved Amount = quoted cost, Status Changed On = `utcNow()`; add Status History row (Changed By = approver).
 - **No:** Update TV Request → Status = Funding Rejected, Reason = approver comments; add Status History row.
11. F02 then sends FUNDING_APPROVED / FUNDING_REJECTED automatically.

Note: a flow run can wait at most **30 days**. F05 sends reminders to the CCO while the approval is pending. If 30 days pass, the run ends; F05 flags it to the APPG owner.

Sample: quoted 39,500 ≤ 50,000 → only Maria Lopez approves → status Funding Approved. Quoted 62,000 → Maria approves, then James Wilson approves → Funding Approved.

F05 – Daily Reminders, Escalations and Auto-Cancellation

Trigger: Recurrence – every 1 day at 08:00, time zone = LAM local, on weekdays.

Steps:

1. Read settings: ReminderAfterDays (3), EscalationAfterDays (5), AutoCancelAfterDays (30).

2. **List rows** – TV Requests waiting for an action:

```
(tvr_status eq 100000001 or tvr_status eq 100000002 or tvr_status eq 100000003
or tvr_status eq 100000005 or tvr_status eq 100000006 or tvr_status eq 100000007)
and tvr_statuschangedon le @{addDays(utcNow(), mul(-1, int(variables('ReminderDays'))))}
```

3. **Apply to each** request:

- Compose **DaysWaiting** = `div(sub(ticks(utcNow()), ticks(items('Apply_to_each')?['tvr_statuschangedon'])), 864000000000)`
- Decide **who must act**: More Info Required → requester; Pending Funding → CCO; other statuses → APPG owner (or APPG team if no owner).
- **Condition A – auto-cancel**: status = More Info Required **and** DaysWaiting ≥ 30 → Update row: Status = Cancelled, Reason = "Auto-cancelled: no response for 30 days"; add Status History (Changed By = system). (F02 sends the CANCELLED email.)
- **Condition B – escalate**: DaysWaiting ≥ 5 → Office 365 Users **Get manager (V2)** of the person who must act → send ESCALATION email to manager, CC the person.
- **Else – reminder**: send REMINDER email to the person who must act.

4. **List rows** – Milestones where `tvr_plannedend lt @{utcNow()}` and status ne Completed → reminder to APPG owner ("Milestone overdue").

5. **List rows** – TV Requests In Execution where Expected Completion < today → email APPG owner.

Sample run (12-Oct-2026):

TVID	Status	Days waiting	Action
TV-2026-00018	Submitted	3	Reminder to APPG team
TV-2026-00016	More Info Required	6	Escalation to Priya's manager
TV-2026-00009	More Info Required	31	Auto-cancelled
TV-2026-00012	Pending Funding Approval	4	Reminder to CCO

Working days only? (ASSUMPTION: calendar days). If the business wants working days, keep a *Holiday* table and count only Mon–Fri non-holiday days.

F06 – Generate PDF and Store

Make two flows:

- **F06a – Build Request PDF (child flow)** – trigger *Manually trigger a flow* with input `TVID`.
- **F06 – Export PDF (from app)** – trigger *Power Apps (V2)* with input `TVID` → **Run a Child Flow F06a** → **Respond to a PowerApp or flow** with `fileurl`.

F06a steps:

1. List rows TV Requests `tvr_tvid eq '{@triggerBody()['text']}'` (expand requester, cost center).
2. List rows Feasibility Options, Funding Approvals, Milestones for this request.
3. **Create HTML table** for options / approvals / milestones.
4. **Compose** full HTML:

```
<html><body style="font-family:Segoe UI">
<h1>TV Request Summary – TV-2026-00015</h1>
<p><b>Title:</b> Plasma etch uniformity test ... <b>Status:</b> Closed</p>
<h2>Feasibility Options</h2> @body('Create_HTML_table_options')
<h2>Funding Approvals</h2> @body('Create_HTML_table_approvals')
<h2>Milestones</h2> @body('Create_HTML_table_milestones')
<h2>Closure</h2><p>Outcome: ... Lessons learned: ...</p>
<p>Generated on @formatDateTime(utcNow(),'dd-MMM-yyyy HH:mm') UTC</p>
</body></html>
```

5. OneDrive for Business **Create file** `/TVRMS-Temp/{TVID}.html` with the HTML.
6. OneDrive **Convert file** (by Id) → Target type **PDF**.
7. SharePoint **Create file** → folder `/@{TVID}/06-Final Report & Closure` (or `01-Request` if not closed), name `@{TVID}_Summary_@formatDateTime(utcNow(),'yyyyMMdd').pdf`.
8. Add a Request Document row (Type = Approval Evidence / Final Report).
9. OneDrive **Delete file** (temp HTML).
10. **Respond to a PowerApp or flow:** `fileurl` = SharePoint file link.

(Option: if LAM has a Word template, use *Populate a Microsoft Word template* (premium) instead of HTML.)

F07 – On Request Closed (Completion Notification)

Trigger: TV Requests modified, Select `tvr_status`, Filter `tvr_status eq 100000013` (Closed).

Steps:

1. **Run a Child Flow** F06a → closure summary PDF saved in `06-Final Report & Closure`.
2. Find the final report link (List rows Request Documents, Type = Final Report, top 1).
3. Send templated email **COMPLETED** → requester, APPG owner, CCO, stakeholders, with links to the final report and the summary PDF ("Report Available Notification").
4. If Request Type = Recurring → send **NEXT_OCCURRENCE** reminder to requester: "Your next Quarterly TV request is due. Use 'Repeat' to create it."

F08 – Error handling pattern (Exception Remediation)

Put every flow's steps inside **Scope – Try**. Then add **Scope – Catch** with *Configure run after* = **has failed, has timed out**:

1. Add a new row – Error Logs:

- Flow Name: `@{workflow()?['tags']?['flowDisplayName']}`
- Error: `@{string(result('Try'))}`
- Run URL:

```
concat('https://make.powerautomate.com/environments/', workflow()?['tags']?['environmentName'],
'/flows/', workflow()?['name'], '/runs/', workflow()?['run']?['name'])
```

- TV Request: the trigger row id.

2. Send email to env var `tvr_AdminEmail`: "TVRMS flow failed – {Flow} – {TVID}" + Run URL.

3. **Terminate** – Status: Failed (so the run shows red in history).

Admin opens **Monitoring** → **Open Errors** in the model-driven app, fixes the cause, and **resubmits** the failed run from flow run history. Then sets Error Log Status = Fixed.

Also set **Retry policy** (Settings of each action) for SharePoint/Outlook actions: *Exponential, 4 retries*.

F09 – Upload Document (called from the app)

Trigger: Power Apps (V2) with inputs:

Input	Type	Sample
TVID	Text	TV-2026-00015
DocType	Text	Technical Specification
Folder	Text	02-Technical Documents
file	File	Etch_Test_Spec_v1.pdf

Steps:

1. SharePoint **Create file**

- Folder path: `/@{parameters('SharePoint Library (tvr_SPLibrary)')}/@{triggerBody()['text']}/@{triggerBody()['text_2']}`
- File name: `@{triggerBody()?['file']?['name']}`
- File content: `@{triggerBody()?['file']?['contentBytes']}` (If the folder does not exist yet, SharePoint creates it. Same file name again = new version.)

2. SharePoint **Update file properties** (Id = `outputs('Create_file')?['body/ItemId']`): TVID, DocumentType, Phase.

3. **List rows** TV Requests where `tvr_tvid eq '@{triggerBody()['text']}'`.

4. **Add a new row** – Request Documents: File Name, Document Type, Folder, SharePoint URL, TV Request.

5. **Respond to a PowerApp or flow** – `fileurl`.

App call (from doc 06):

```
'F09-UploadDocument'.Run(varNew.TVID, "Technical Specification", "02-Technical Documents",
{ file: { name: f.Name, contentBytes: f.Value } })
```

F10 – Add Stakeholder (optional, from app)

Power Apps (V2) trigger: TVID, StakeholderEmail → Get user → `AddUserToRecordTeam` (template "TVR Request Team") → email STAKEHOLDER_ADDED.

Flow testing checklist

Flow	How to test	Pass when
F01	Submit a request	6 folders created, 2 emails, URL saved
F02	Change status to More Info Required	Requester gets MORE_INFO email with comment
F04	Send for funding (amount 39,500)	CCO gets approval in Outlook/Teams; decision saved
F04	Amount 62,000	Two approvals in sequence
F05	Set Status Changed On to 31 days ago on a More Info request; run flow manually	Request auto-cancelled
F06	Click Export PDF	PDF opens; file in SharePoint
F07	Close request	Completion email with links
F08	Temporarily rename SharePoint library in env var; submit	Error Log row + admin email
F09	Upload 2 files	Both in SharePoint + 2 Request Document rows

09 – Power BI Dashboard and Reports

Architecture slide – Dashboard and Reporting Layer: **Request Status / Cycle Time & Aging / Cost Tracking / Utilization & Performance / Exception Analysis / Executive Reporting.**

1. Connect to data

1. Open **Power BI Desktop** → **Get data** → **Dataverse** → sign in → choose environment **LAM-TVRMS-PROD** (use DEV while building).
2. Select tables: **tvr_tvrequest**, **tvr_feasibilityoption**, **tvr_fundingapproval**, **tvr_milestone**, **tvr_riskissue**, **tvr_costentry**, **tvr_statushistory**, **tvr_department**, **tvr_costcenter**, **tvr_location**, **tvr_researchinstitution**.
3. Mode: **Import** (fast, refresh 4–8 times a day). Use DirectQuery only if live data is a must.
4. In **Power Query**: keep only needed columns; rename them to friendly names; replace choice numbers with labels (the Dataverse connector gives **...name** columns for choices).
5. Add a **Date table** (DAX):

```
Date = ADDCOLUMNS(CALENDAR(DATE(2026,1,1), DATE(2030,12,31)),
  "Year", YEAR([Date]), "Month", FORMAT([Date],"MMM yyyy"),
  "MonthNo", YEAR([Date])*100+MONTH([Date]), "Quarter", "Q" & QUARTER([Date]))
```

6. **Relationships**: TV Request → Department, Cost Center, Location (many-to-one); Feasibility/Funding/Milestone/Risk/Cost/StatusHistory → TV Request (many-to-one); Date[Date] → TV Request[Submitted On (date)].

2. DAX measures (copy into a "_Measures" table)

```
Total Requests      = COUNTROWS('TV Request')
Open Requests       = CALCULATE([Total Requests],
  NOT 'TV Request'[Request Status] IN {"Closed","Cancelled","Rejected","Funding Rejected"})
Closed Requests     = CALCULATE([Total Requests], 'TV Request'[Request Status] = "Closed")
Rejected Requests   = CALCULATE([Total Requests], 'TV Request'[Request Status] IN {"Rejected","Funding Rejected"})
Approval Rate %     = DIVIDE(CALCULATE([Total Requests], NOT ISBLANK('TV Request'[Approved Amount])),
  CALCULATE([Total Requests], 'TV Request'[Request Status] IN {"Funding Approved","Funding Rejected","In Execution","On Hold","Execution Completed","Closed"}))

Avg Cycle Time (days) = AVERAGEX(FILTER('TV Request', NOT ISBLANK('TV Request'[Closed On])),
  DATEDIFF('TV Request'[Submitted On], 'TV Request'[Closed On], DAY))

Age (days) request  = DATEDIFF(MAX('TV Request'[Submitted On]), TODAY(), DAY)  -- use in table visual per request

Days In Current Status = DATEDIFF(MAX('TV Request'[Status Changed On]), NOW(), DAY)

Approved Budget      = SUM('TV Request'[Approved Amount])
Actual Cost          = SUM('Cost Entry'[Amount])
Cost Variance        = [Approved Budget] - [Actual Cost]
Budget Utilization % = DIVIDE([Actual Cost], [Approved Budget])

On-Time Completion % = DIVIDE(
  CALCULATE([Closed Requests], FILTER('TV Request', 'TV Request'[Closed On] <= 'TV Request'[Required By Date])),
  [Closed Requests])
```

```

Open Issues      = CALCULATE(COUNTROWS('Risk Issue'), 'Risk Issue'[Status] <> "Resolved")
High Severity Issues= CALCULATE([Open Issues], 'Risk Issue'[Severity] = "High")
Overdue Milestones = CALCULATE(COUNTROWS('Milestone'),
                                'Milestone'[Planned End] < TODAY(), 'Milestone'[Status] <> "Completed")

```

Aging bucket (calculated column on TV Request):

```

Aging Bucket =
VAR d = DATEDIFF('TV Request'[Submitted On], IF(ISBLANK('TV Request'[Closed On]), TODAY(), 'TV
Request'[Closed On]), DAY)
RETURN SWITCH(TRUE(), d <= 15, "0-15 days", d <= 30, "16-30 days", d <= 60, "31-60 days", "60+ days")

```

Time spent in each phase (from Status History) – calculated column on Status History:

```

Hours In Status =
VAR thisReq = 'Status History'[TV Request]
VAR thisTime = 'Status History'[Changed On]
VAR nextTime = CALCULATE(MIN('Status History'[Changed On]),
    FILTER(ALL('Status History'), 'Status History'[TV Request] = thisReq && 'Status
History'[Changed On] > thisTime))
RETURN DATEDIFF(thisTime, IF(ISBLANK(nextTime), NOW(), nextTime), HOUR)

```

Then average of **Hours In Status** by **To Status** = bottleneck chart.

3. Report pages

Page	Visuals	Answers the question
1. Executive Summary	KPI cards (Total, Open, Closed, Avg Cycle Time, Approved Budget, Utilization %), requests by month (column), status funnel	"How are we doing overall?"
2. Request Status	Bar by status, donut by phase, table of open requests with TVID / owner / days in status, slicers (Department, Request Type, Priority, Date)	"Where is every request now?"
3. Cycle Time & Aging	Aging bucket bar, avg cycle time by department, hours in each status (bottleneck), on-time %	"Where do we lose time?"
4. Cost Tracking	Approved vs Actual by request (clustered bar), cost by category (pie), by cost center (matrix), variance	"Are we within budget?"
5. Utilization & Performance	Requests per research institution, avg quoted cost & weeks per institution, budget utilization %, reuse recommendation counts	"Which institutions perform best?"
6. Exception Analysis	Open issues by severity, overdue milestones list, auto-cancelled & rejected requests with reasons, failed notifications	"What is going wrong?"

4. Sample numbers (for testing visuals with sample data)

KPI	Expected with /sample-data
Total Requests	8
Open Requests	5
Closed Requests	1
Rejected / Cancelled	2
Approved Budget	39,500 + 28,000 + 15,000 = 82,500

5. Security in Power BI

- Dataverse connector **respects Dataverse security of the person who refreshes**. With Import mode the service account refreshes → everyone sees everything in the report. So: share the **Executive** report only with APPG + management (`SG-TVRMS-APPG`, leadership group).
- If requesters need a "My requests" report, add **Row-Level Security**: role "Requester" with filter on TV Request: `[Requester Email] = USERPRINCIPALNAME()`.

6. Publish

1. Publish to workspace `LAM TVRMS` (Power BI service).
2. Dataset settings → credentials (OAuth, service account) → **Scheduled refresh** 4× per day.
3. Create an **App** in the workspace and share with the right groups.
4. (Optional) Embed the report in the Canvas app with the *Power BI tile* control, or add it as a dashboard page in the Model-Driven app.

10 – Testing and UAT

Slide 6: "Validation of implemented functionality · UAT would be performed with limited users · UAT feedback incorporation · Issues remediation." Slide 8: "UAT will be performed by business users."

1. Types of testing

Type	Who	Where	When
Unit testing (each screen / flow alone)	Developer	DEV	During each sprint
System / integration testing (full process)	Developer + QA	TEST	End of each sprint
Security testing (each role)	Developer	TEST	Sprint 2
UAT	3–5 business users (ASSUMPTION)	TEST	Week 6–7
Smoke test after deployment	Developer + 1 key user	PROD	Go-live day

2. Test users (create in TEST, one per role)

Test user	Role	Department / CC
test.requester1@lam.com (Priya)	TVR Requester	Process Engineering
test.requester2@lam.com (John)	TVR Requester	Metrology
test.appg@lam.com (David)	TVR APPG Reviewer	–
test.cco@lam.com (Maria)	TVR Cost Center Owner	CC-4100
test.ccomgr@lam.com (James)	Manager of Maria	–
test.stakeholder@lam.com (Tom)	TVR Stakeholder	–
test.admin@lam.com	TVR Admin	–

3. Test cases (with sample data)

ID	Phase	Scenario	Steps (sample data)	Expected result
TC-01	Initiation	Mandatory fields	Priya clicks Submit with Title empty	Error "Please fill all fields marked with *"; nothing saved
TC-02	Initiation	Technical document mandatory	Fill all fields, no attachment, Submit	Error "attach at least one technical specification"
TC-03	Initiation	Past date	Required By = yesterday	Error about future date
TC-04	Initiation	Happy path New	Title "Plasma etch uniformity test", New, PE, Fremont, CC-4100, High, 15-Dec-2026, 42,000, attach Etch_Test_Spec_v1.pdf, Submit	TVID TV-2026-000xx created, status Submitted, 6 SP folders, file in 02-Technical Documents, ack email to Priya, email to APPG
TC-05	Initiation	Repeat request	Type = Repeat, select TV-2026-00003	Fields pre-filled; Previous Request saved
TC-06	Initiation	Recurring without frequency	Type = Recurring, no frequency	Error, cannot submit
TC-07	Initiation	Save Draft	Fill Title only → Save Draft	Saved as Draft; no email; no folder

ID	Phase	Scenario	Steps (sample data)	Expected result
TC-08	APPG Review	Start review & assign owner	David opens TV, Start Review, owner = David	Status Under APPG Review; email REVIEW_STARTED to Priya; history row
TC-09	APPG Review	More info	David: comment "Please attach the wafer map" → Request More Info	Status More Info Required; Priya gets email with comment
TC-10	APPG Review	More info without comment	Click Request More Info with empty comment	Error; status unchanged
TC-11	APPG Review	Requester responds	Priya uploads Wafer_Map.xlsx, Resubmit	Status Under APPG Review; APPG notified
TC-12	APPG Review	Reject	David rejects with reason "Duplicate of TV-2026-00011"	Status Rejected; reason saved; Priya email
TC-13	Alignment	Add 3 options	Options A/B/C (see doc 06 §10)	3 rows; comparison visible
TC-14	Alignment	Select option	Select Option A	Only A is selected; Estimated budget = 39,500
TC-15	Alignment	Move to Alignment 2 without option	Click Confirm option, none selected	Button disabled / error
TC-16	Funding	Normal approval	Send for Funding (39,500) → Maria approves "OK from Q4 budget"	Funding row Approved; status Funding Approved; Approved Amount 39,500; emails
TC-17	Funding	High value	Amount 62,000 → Maria approves → James approves	2 funding rows (L1, L2); Funding Approved
TC-18	Funding	Rejection	Maria rejects "No budget this quarter"	Status Funding Rejected; Priya gets reason
TC-19	Execution	Milestones & progress	Add 4 milestones; set 2 to 100%	Overall Progress = 50%
TC-20	Execution	Cost tracking	Add costs 18,000 + 12,000	Actual 30,000; Utilization 76%
TC-21	Execution	High issue	Add Issue "Lab tool down", severity High	EXCEPTION_RAISED email
TC-22	Closure	Close without final report	Click Close	Error "upload the final report first"
TC-23	Closure	Close	Fill outcome, lessons, reuse; upload Final_Report.pdf; Close	Status Closed; summary PDF in 06 folder; completion email to all
TC-24	Automation	Reminder	Request Submitted 3 days ago (set date in TEST) → run F05	Reminder email to APPG
TC-25	Automation	Escalation	More Info Required 6 days	Email to requester's manager
TC-26	Automation	Auto-cancel	More Info Required 31 days	Status Cancelled, reason "Auto-cancelled..."
TC-27	Security	Requester isolation	John opens app	Cannot see Priya's requests
TC-28	Security	CCO view	Maria opens link from approval email	Can view TV-2026-00015 read-only
TC-29	Export	PDF	Click Export to PDF	PDF opens; saved in SharePoint
TC-30	Audit	History	Open Audit History in admin app	Every field change with user & time
TC-31	Admin	Change setting	ReminderAfterDays = 2	Next F05 run uses 2 days
TC-32	NFR	Browser	Open app in Microsoft Edge & Chrome	Works the same
TC-33	Reports	Dashboard	Refresh Power BI after tests	Counts match Dataverse

4. UAT process

1. **Prepare:** load master data + sample data into TEST; give testers the test-case sheet (export the table above to Excel with columns *Actual result*, *Pass/Fail*, *Comments*).
2. **Kick-off meeting (30 min):** show the app, explain how to log issues.
3. **Testers execute** for 3–5 days. Log defects in a list (Excel, Azure DevOps, or a SharePoint list):

Defect ID	Test case	Description	Severity	Status	Fixed in
D-001	TC-09	Email shows {Comments} instead of the comment	High	Fixed	Build 1.0.0.5
D-002	TC-19	Progress % not refreshing on screen	Medium	Fixed	1.0.0.5
D-003	–	Want Department dropdown sorted A-Z	Low (enhancement)	Done	1.0.0.6

4. Fix → redeploy to TEST → retest.

5. **UAT sign-off** email from business owner (slide 6: "Key stakeholders would need to approve and provide sign off for deployment").

Severity rules: **Critical** – blocks process (must fix); **High** – wrong result (must fix); **Medium** – workaround exists (fix before go-live if possible); **Low** – cosmetic. New ideas = **change request** (fixed price, slide 10).

11 – Deployment, Go-Live and Hypercare

Slide 6: Deployment = *key stakeholders approve & sign off, Go-Live preparation, Go-Live*. Hypercare = *post rollout support, handover to application support team*. Deliverables: **Signed-off Power Platform Applications** and **Handover to Support**.

1. ALM – how the solution moves

```

DEV (unmanaged solution) —export managed—> TEST (managed) —same file—> PROD (managed)
      |
      └─ export unmanaged + unpack —> this Git repository (folder /solution)
  
```

Never build directly in TEST or PROD.

Option A – Power Platform Pipelines (easiest, built-in)

1. Admin installs **Power Platform Pipelines** in a host environment.
2. Create pipeline **TVRMS Pipeline**: Stage 1 = TEST, Stage 2 = PROD (source = DEV).
3. In DEV: Solutions → TVRMS → **Pipelines** → **Deploy here**. Fill environment variable values for the target and select connections. Approvals can be required for PROD.

Option B – Manual export/import

1. DEV → Solutions → TVRMS → **Publish all customizations**.
2. Increase version, e.g. `1.0.0.5` → `1.0.1.0`.
3. **Export → Managed** → you get `TVRMS_1_0_1_0_managed.zip`.
4. TEST → Solutions → **Import** → choose zip → set connection references (service account connections) and environment variable values (TEST SharePoint URL, etc.) → Import.
5. After testing the **same zip** goes to PROD.

Keep the source in Git (this repository)

```

# one-time
pac auth create --environment https://lam-tvrms-dev.crm.dynamics.com
# every time you finish a feature
pac solution export --name TVRMS --path ./out/TVRMS.zip --managed false
pac solution unpack --zipfile ./out/TVRMS.zip --folder ./solution --packagetype Unmanaged
git add solution && git commit -m "TVRMS 1.0.1.0 - funding approval flow" && git push
  
```

2. What is NOT inside the solution (do separately in each environment)

Item	How
Master data (departments, cost centers, ...)	Import CSV from LAM (slide 8: customer provides)
App Settings + Email Templates rows	Import <code>sample-data/app_settings.csv</code> and <code>email_templates.csv</code> (or Configuration Migration tool)
SharePoint site, library, columns, versioning, retention label	Create manually (doc 04) or PnP script
Entra ID groups & Dataverse group teams	IT + admin center (doc 05)
Auditing settings	Admin center (doc 02)
Power BI report	Publish from Desktop, change data source to PROD

Item	How
Sharing app with groups	Share after import
Turning flows ON	Check every flow is On after import

3. Go-Live checklist (Week 7)

1 week before

- ☐ UAT sign-off received (email saved in project folder)
- ☐ PROD environment, licenses assigned to all users
- ☐ PROD SharePoint site + library + retention label ready
- ☐ Master data from LAM received & validated (no blanks in Cost Center Owner!)
- ☐ Email templates approved by business
- ☐ Go-live communication drafted (by LAM – change management is out of scope)

Go-live day

- ☐ Import managed solution to PROD (version noted, e.g. 1.0.2.0)
- ☐ Set environment variables & connections (service account)
- ☐ Load master data, settings, templates
- ☐ Turn on all flows; check owners = service account
- ☐ Share Canvas app (users) and Model-driven app (admins)
- ☐ Publish Power BI, set refresh, share app
- ☐ **Smoke test** in PROD: create 1 test request → submit → check folder + email → cancel it (mark it "TEST – ignore")
- ☐ Send go-live email with app link

Rollback plan: if a critical issue appears, turn off flows, inform users to use email temporarily, fix in DEV, redeploy. (Managed solutions can be upgraded quickly.)

4. Hypercare (Week 8–9, about 2 weeks)

Activity	Detail
Daily check	Flow run history (failed runs), Error Log, Failed Notifications view
Daily 15-min call	With APPG lead: issues, questions
Issue log	Same format as UAT defects
Quick fixes	Fix in DEV → TEST → PROD (small releases)
Usage check	How many requests submitted; any stuck requests

5. Handover to the Application Support Team (BAU)

Give them a **handover pack**:

1. This repository (all docs + exported solution).
2. Architecture diagram and data model (docs 03).
3. List of flows with purpose, trigger, owner (doc 08).

4. Security roles & how to add users (doc 05) – "add person to Entra group X".

5. **Runbook** – common support tasks:

Problem	Fix
User cannot open app	Check license + member of <code>SG-TVRMS-Users</code>
User cannot see a request	Check role / access team / ownership
Email not received	Model-driven app → Failed Notifications → error; check flow run
Folder not created	Error Log → resubmit F01 run
CCO left the company	Admin app → Cost Center → change owner; reassign pending approval (cancel approval & re-send)
Change reminder days	App Settings table
Flow connection expired	Fix connection of service account; all flows use connection references

6. Credentials ownership: service account managed by LAM IT (never a personal account).

7. Knowledge transfer session (1–2 hours, recorded).

12 – Project Plan (9 weeks, from slide "Execution Timeline")

Week	Phase (slide)	Main work	Output
W1	Plan for Success	Kick-off; workshops (process, data & documents, reports); send question list (doc 00); get access & licenses; create environments	Workshop notes, answers
W2	Plan for Success	Write Solution Blueprint (data model, statuses, screens, flows, emails, KPIs, security); validate plan; submit for approval	Deliverable: Solution Report (signed-off)
W3	Sprint 1	Dataverse tables, choices, relationships, security roles; master data import; SharePoint site; Canvas app shell + Home + New Request wizard; F09 upload; F01 submit	Request intake working
W4	Sprint 1	Request Detail, APPG Queue, review actions (more info / reject / assign), comments, documents; F02 notifications; email templates; Sprint 1 demo	Demo 1 to business
W5	Sprint 2	Feasibility screen, Internal Alignment 1 & 2, Funding screen, F04 approval routing, access team; Execution screen (milestones, risks, costs)	Full process until execution
W6	Sprint 2	Closure, F06 PDF, F07 completion, F05 reminders/escalations/auto-cancel, F08 error handling, Model-driven admin app, Power BI report; system test; UAT starts; Sprint 2 demo	UAT build
W7	Deploy	UAT fixes, UAT sign-off, PROD setup, Go-Live ★	Deliverable: Signed-off Power Platform Applications
W8	Hypercare	Post-rollout support, fixes	Issue log
W9	Hypercare	Support, knowledge transfer, handover to BAU team	Deliverable: Handover to Support

Sprint backlog (user stories with acceptance criteria)

Sprint 1 (W3–W4)

ID	User story	Acceptance criteria	Points
US-01	As an admin I want master data tables so that forms have dropdowns	Departments, CC, Locations, RIs created + imported	3
US-02	As a requester I want to create a TV request (New/Repeat/One-time/Recurring)	Wizard with 3 steps; Repeat copies data; TVID auto-generated	8
US-03	As a requester I want the system to check mandatory fields & technical document	TC-01..03, TC-06 pass	3
US-04	As a requester I want to upload documents	Files go to TVID folder; metadata row created	5
US-05	As the business I want a TVID workspace created automatically	6 folders on submit	3
US-06	As a requester I want an acknowledgement email	Email within 2 min with link	2
US-07	As APPG I want a queue of requests and to assign an owner	Queue sorted by waiting days	5
US-08	As APPG I want to request more information / reject with comments	Status + email + history	5
US-09	As a requester I want to see my requests and status	Home screen counters + list	3
US-10	As the business I want role-based access	TC-27 passes	5

Sprint 2 (W5–W6)

ID	User story	Acceptance criteria	Points
US-11	As APPG I want to compare feasibility options	Add/compare/select options	5
US-12	As APPG I want to run internal alignment 1 and 2	Status moves; alignment emails	3
US-13	As APPG I want funding approval routed to the CCO	TC-16..18 pass	8
US-14	As APPG I want to track milestones, risks, costs	TC-19..21 pass	8
US-15	As APPG I want to close a request with outcome & lessons learned	TC-22..23 pass	5
US-16	As the business I want reminders, escalations, auto-cancel	TC-24..26 pass	5
US-17	As a user I want to export a request to PDF	TC-29 passes	3
US-18	As an admin I want an admin app for master data and logs	Model-driven app with views	3
US-19	As management I want a Power BI dashboard	6 pages, numbers match	8
US-20	As support I want failed flows logged	Error Log + email	2

Team (ASSUMPTION – typical for this size)

Role	Allocation
Project Manager / Business Analyst	50%
Power Platform Developer (Apps + Flows)	100%
Power Platform Developer / Power BI	100% (W3–W7)
QA / Tester	50% (W4–W7)

Risks to watch

Risk	Impact	Action
Late master data from LAM	Can't test properly	Use sample data; ask in W1
Licenses not ready	Blocks build/testing	Raise on Day 1 (slide 8 dependency)
Requirements change after blueprint	Delay, cost	Change request process
SMEs not available for workshops / UAT	Delay	Book calendars in W1

13 – End-to-End Use Case (one request from start to finish)

Story: Priya Sharma, a process engineer in Fremont, needs a plasma etch uniformity test because customer X reported 4% non-uniformity. Let us follow her request **TV-2026-00015** through all 6 phases and see what the system does at each step.

Phase 1 – Request Initiation (12-Oct-2026)

Priya opens the app → **New Request:**

Field	Value
Title	Plasma etch uniformity test on 300mm wafers
Request Type	New
Department	Process Engineering
Location	Fremont, CA
Cost Center	CC-4100 Etch R&D
Priority	High
Required By	15-Dec-2026
Estimated Budget	USD 42,000
Business Justification	Customer X reports 4% within-wafer non-uniformity on 3nm etch. Need to qualify new recipe.
Technical Description	25 wafers, 3 recipes, CD-SEM measurement at 49 points per wafer.
Expected Outcome	Uniformity below 2%
Attachment	Etch_Test_Spec_v1.pdf

She clicks **Submit**.

System does:

1. App validates fields + attachment  → saves the record → TVID **TV-2026-00015**.
2. F09 uploads the PDF to `TV Requests/TV-2026-00015/02-Technical Documents`.
3. Status → **Submitted**; history row "Draft → Submitted".
4. F01 creates the 6 folders, saves the folder URL, emails Priya (**acknowledgement**) and the APPG mailbox (**new request**).

Phase 2 – APPG Review (13–16 Oct)

- **13-Oct, David (APPG)** opens the APPG Queue → TV-2026-00015 → **Start Review**, APPG Owner = David. → Status **Under APPG Review**; Priya gets "review started" email.
- David notices the wafer map is missing → comment *"Please attach the wafer map for the 25 wafers."* → **Request More Info**. → Status **More Info Required**; Priya gets email with the comment.
- **15-Oct, Priya** uploads `Wafer_Map.xlsx`, writes *"Wafer map attached."* → **Resubmit**. → Status **Under APPG Review**; David is notified. (*If Priya had not answered: reminder on day 3, escalation to her manager on day 5, auto-cancel on day 30 – flow F05.*)
- **16-Oct, David** is satisfied → **Move to Internal Alignment**.

Phase 3 – Internal Alignment (17 Oct – 31 Oct)

1st Internal Alignment – David asks 3 institutions for quotes and enters options:

Option	Institution	Capability	Quote (USD)	Weeks	Constraint
A	Stanford Nanofab Lab	5	39,500	10	Lab free from Nov
B	Fraunhofer IPMS	4	35,000	16	Shipping to Germany
C	Internal Lab Fremont	3	22,000	20	Tool booked till Jan

Quotes uploaded to **03-Feasibility & Quotes**. Requester gets an **alignment update** email.

2nd Internal Alignment – meeting with Priya and program manager Tom: Option **A** chosen because only it meets the 15-Dec date. David clicks **Select this option** → **Confirm Option** (status Internal Alignment 2), adds Tom as stakeholder, then **Send for Funding Approval**.

Phase 4 – Funding Approval (2–3 Nov)

F04 runs:

1. Cost center CC-4100 → owner **Maria Lopez**.
2. Amount 39,500 ≤ limit 50,000 → only one approval level.
3. Maria is added to the request's access team (can now view it).
4. Funding Approval row created (Pending). Approval appears in Maria's Outlook / Teams.

3-Nov, Maria clicks **Approve**, comment *"Approved from Q4 budget."*

→ Funding Approval row = Approved; Request **Funding Approved**, Approved Amount 39,500; Priya and David get **funding approved** email.

(If Maria had rejected with "No budget this quarter" → status Funding Rejected, Priya gets the reason; request ends.)

Phase 5 – Execution (10 Nov – 28 Feb)

David clicks **Start Execution** → status **In Execution**. He adds milestones:

#	Milestone	Planned	Actual	%
1	Wafer preparation	10–24 Nov	12–26 Nov	100
2	Etch runs (3 recipes)	25 Nov–15 Dec	27 Nov–20 Dec	100
3	CD-SEM measurement	16 Dec–20 Jan	21 Dec–25 Jan	100
4	Final report	21 Jan–15 Feb	26 Jan–20 Feb	100

- **5-Dec:** issue "Lab tool down for maintenance", severity High → **exception email** to David and Priya. Mitigation: move run to 2nd tool → Resolved 9-Dec.
- Costs entered: Lab fee 18,000 (Nov) + Lab fee 12,000 (Dec) + Materials 5,300 + Shipping 2,500 = **37,800** → Utilization **96%** of 39,500.
- Expected completion updated to 28-Feb. Status updates emailed to Priya and Tom.
- 28-Feb: David clicks **Complete Execution** → status Execution Completed.

Phase 6 – Closure (5 Mar 2027)

David uploads `Final_Report.pdf` to `06-Final Report & Closure` and fills:

Field	Value
Final Outcome	Recipe B improved uniformity to 1.6% (target < 2%) 
Lessons Learned	Book external lab 6 weeks early; keep backup tool option.
Reuse Recommendation	Reuse with changes

Clicks **Close Request** → status **Closed**, Closed On 5-Mar-2027.

F07: builds summary PDF (`TV-2026-00015_Summary_20270305.pdf`) into SharePoint, and emails Priya, David, Maria and Tom: **"TV-2026-00015 completed – final report available"** with links.

What management sees in Power BI

- Cycle time for TV-2026-00015 = 12-Oct-2026 → 5-Mar-2027 = **144 days**.
- Cost: approved 39,500, actual 37,800, variance +1,700.
- Bottleneck: most time spent in *In Execution* (expected), 2 days lost in *More Info Required*.
- Stanford Nanofab: 1 request, 100% on budget.

Full audit trail (Status History for TV-2026-00015)

When	From → To	By
12-Oct 10:05	Draft → Submitted	Priya
13-Oct 09:12	Submitted → Under APPG Review	David
13-Oct 09:40	Under APPG Review → More Info Required	David
15-Oct 14:20	More Info Required → Under APPG Review	Priya
16-Oct 11:00	Under APPG Review → Internal Alignment 1	David
31-Oct 16:30	Internal Alignment 1 → Internal Alignment 2	David
02-Nov 09:00	Internal Alignment 2 → Pending Funding Approval	David
03-Nov 15:45	Pending Funding Approval → Funding Approved	Maria (via flow)
10-Nov 08:30	Funding Approved → In Execution	David
28-Feb 17:00	In Execution → Execution Completed	David
05-Mar 10:10	Execution Completed → Closed	David

Sample data (for DEV / TEST only)

Fake data that matches the examples in the docs. Import in this order (lookups need parents first):

1. `departments.csv`
2. `locations.csv`
3. `research_institutions.csv`
4. `cost_centers.csv`
5. `app_settings.csv`
6. `email_templates.csv`
7. `tv_requests.csv`
8. `feasibility_options.csv`
9. `milestones.csv`
10. `cost_entries.csv`

Notes:

- People are given by **email**. These users must exist in the environment (from Entra ID). Replace the emails with your real test users before import.
- In `tv_requests.csv` the TVID values are given so the examples line up. In real use the TVID is an **autonumber**; when importing, either map TVID to the autonumber column (allowed on import) or let it generate new numbers.
- Choice columns (Request Type, Request Status, Priority ...) must use the exact labels you created.
- Expected Power BI numbers with this data: Total 8, Open 5, Closed 1, Rejected/Cancelled 2, Approved budget 82,500.
- **Never** load this sample data into PROD.

Sample data – departments.csv

Loads into table: Department (`tvr_department`) · Import order: 1 of 10 · Rows: 4 · Columns: 4

Fake test data for DEV / TEST environments only – never load into PROD. People are given by e-mail and must exist in the environment first.

#	Name	Department Code	Department Head (email)	Is Active
1	Process Engineering	PE	ravi.kumar@lam.com	Yes
2	Metrology	MET	susan.lee@lam.com	Yes
3	Deposition R&D	DEP	kevin.brown@lam.com	Yes
4	Product Engineering	PRD	anna.white@lam.com	Yes

Sample data – locations.csv

Loads into table: Location (`tvr_location`) · Import order: 2 of 10 · Rows: 4 · Columns: 3

Fake test data for DEV / TEST environments only – never load into PROD. People are given by e-mail and must exist in the environment first.

#	Name	Country	Is Active
1	Fremont, CA	USA	Yes
2	Tualatin, OR	USA	Yes
3	Bangalore	India	Yes
4	Villach	Austria	Yes

Sample data – research_institutions.csv

Loads into table: Research Institution / Vendor (`tvr_researchinstitution`) · **Import order:** 3 of 10 · **Rows:** 4 · **Columns:** 6

Fake test data for DEV / TEST environments only – never load into PROD. People are given by e-mail and must exist in the environment first.

#	Name	Type	Contact Name	Contact Email	Capabilities	Is Active
1	Stanford Nanofab Lab	University	Dr. Alan Green	agreen@stanford.example	Plasma etch, SEM, TEM	Yes
2	Fraunhofer IPMS	Research Lab	Dr. Eva Schmidt	eva.schmidt@ipms.example	Etch, deposition, metrology	Yes
3	Internal Lab Fremont	Internal Lab	Mark Davis	mark.davis@lam.com	Etch, CD-SEM	Yes
4	IMEC	Research Lab	Dr. Pieter Claes	pieter.claes@imec.example	Advanced nodes, EUV	Yes

Sample data – cost_centers.csv

Loads into table: Cost Center (`tvr_costcenter`) · Import order: 4 of 10 · Rows: 3 · Columns: 6

Fake test data for DEV / TEST environments only – never load into PROD. People are given by e-mail and must exist in the environment first.

#	Name	Cost Center Code	Cost Center Owner (email)	Owner Manager (email)	Department	Is Active
1	CC-4100 Etch R&D	4100	maria.lopez@lam.com	james.wilson@lam.com	Process Engineering	Yes
2	CC-4200 Metrology	4200	susan.lee@lam.com	james.wilson@lam.com	Metrology	Yes
3	CC-4300 Deposition R&D	4300	kevin.brown@lam.com	james.wilson@lam.com	Deposition R&D	Yes

Sample data – app_settings.csv

Loads into table: App Setting (`tvr_appsetting`) · Import order: 5 of 10 · Rows: 5 · Columns: 3

Fake test data for DEV / TEST environments only – never load into PROD. People are given by e-mail and must exist in the environment first.

#	Name	Value	Description
1	ReminderAfterDays	3	Send reminder if a pending action is older than this many days
2	EscalationAfterDays	5	Escalate to the manager after this many days
3	AutoCancelAfterDays	30	Auto-cancel a request in More Info Required after this many days
4	HighValueFundingLimit	50000	Above this amount (USD) a second-level funding approval is needed
5	RetentionYears	7	Record retention in years

Sample data – email_templates.csv

Loads into table: Email Template (`tvr_emailtemplate`) · **Import order:** 6 of 10 · **Rows:** 15 · **Columns:** 4

Fake test data for DEV / TEST environments only – never load into PROD. People are given by e-mail and must exist in the environment first.

#	Template Code	Subject	Body (HTML)	Is Active
1	REQ_SUBMITTED	[TVRMS] {TVID} submitted - {Title}	Dear {RequesterName}, Your Test Vehicle request {TVID} has been submitted and is now with the APPG team for review. Open request Regards, TV Request Management System	Yes
2	NEW_REQUEST_APPG	[TVRMS] New request {TVID} waiting for review	Hello APPG team, A new request {TVID} - {Title} was submitted by {RequesterName}. Review now	Yes
3	REVIEW_STARTED	[TVRMS] {TVID} is under APPG review	Dear {RequesterName}, Your request {TVID} is now under review. Open request	Yes
4	MORE_INFO	[TVRMS] Action needed: more information for {TVID}	Dear {RequesterName}, The APPG team needs more information for {TVID}: <blockquote>{Comments}</blockquote> Please update and resubmit. Open request	Yes
5	REJECTED	[TVRMS] {TVID} rejected	Dear {RequesterName}, Your request {TVID} was rejected. Reason: {Comments}	Yes
6	ALIGNMENT_UPDATE	[TVRMS] {TVID} internal alignment update	Request {TVID} is now in status {Status}. {Comments} Open request	Yes
7	FUNDING_APPROVED	[TVRMS] Funding approved for {TVID}	Good news! Funding for {TVID} was approved. Comments: {Comments} Open request	Yes
8	FUNDING_REJECTED	[TVRMS] Funding rejected for {TVID}	Funding for {TVID} was rejected. Reason: {Comments}	Yes
9	STATUS_UPDATE	[TVRMS] {TVID} status changed to {Status}	Request {TVID} is now {Status}. {Comments} Open request	Yes
10	EXCEPTION_RAISED	[TVRMS] High severity issue on {TVID}	A high severity issue was raised on {TVID}: {Comments} Open request	Yes
11	REMINDER	[TVRMS] Reminder: {TVID} is waiting for your action	Request {TVID} has been in status {Status} for {Days} days and needs your action. Open request	Yes
12	ESCALATION	[TVRMS] Escalation: {TVID} pending for {Days} days	Request {TVID} has been waiting for {Days} days for action by {ActionOwner}. Please help to move it forward. Open request	Yes
13	CANCELLED	[TVRMS] {TVID} cancelled	Request {TVID} was cancelled. Reason: {Comments}	Yes

#	Template Code	Subject	Body (HTML)	Is Active
14	COMPLETED	[TVRMS] {TVID} completed - final report available	Request {TVID} - {Title} is closed. Final report: open Summary PDF: open	Yes
15	NEXT_OCCURRENCE	[TVRMS] Next occurrence due for recurring request {TVID}	Your recurring request {TVID} is closed. Please create the next occurrence using Request Type = Repeat.	Yes

Sample data – tv_requests.csv

Loads into table: TV Request (tvr_tvrequest) · Import order: 7 of 10 · Rows: 8 · Columns: 23

Fake test data for DEV / TEST environments only – never load into PROD. People are given by e-mail and must exist in the environment first.

This file has 23 columns, so each row is shown as a card (column name on the left, value on the right).

Row 1 – TV-2026-00001 · Quarterly contamination check	
TVID	TV-2026-00001
Request Title	Quarterly contamination check
Request Type	Recurring
Previous Request	(empty)
Recurrence Frequency	Quarterly
Requester (email)	john.miller@lam.com
Department	Metrology
Location	Fremont, CA
Cost Center	CC-4200 Metrology
Priority	Medium
Required By Date	2026-06-30
Estimated Budget	16000
Request Status	Closed
Current Phase	Closure
APPG Owner (email)	david.chen@lam.com
Approved Amount	15000
Submitted On	2026-03-02 09:00
Status Changed On	2026-06-25 16:00
Closed On	2026-06-25 16:00
Business Justification	Regular particle contamination check
Technical Description	Particle count on 10 wafers
Expected Outcome	Contamination within spec
Rejection / Cancellation Reason	(empty)

Row 2 – TV-2026-00003 · Chamber material stress test	
TVID	TV-2026-00003
Request Title	Chamber material stress test
Request Type	One-time
Previous Request	(empty)
Recurrence Frequency	(empty)
Requester (email)	priya.sharma@lam.com
Department	Process Engineering
Location	Fremont, CA
Cost Center	CC-4100 Etch R&D
Priority	High
Required By Date	2026-11-30
Estimated Budget	30000
Request Status	In Execution
Current Phase	Execution
APPG Owner (email)	david.chen@lam.com
Approved Amount	28000
Submitted On	2026-07-10 10:00
Status Changed On	2026-08-20 09:00
Closed On	(empty)
Business Justification	New chamber liner material
Technical Description	Thermal cycling 500 cycles
Expected Outcome	No cracks after 500 cycles
Rejection / Cancellation Reason	(empty)

Row 3 – TV-2026-00009 · Low-k film adhesion test	
TVID	TV-2026-00009
Request Title	Low-k film adhesion test
Request Type	New
Previous Request	(empty)
Recurrence Frequency	(empty)
Requester (email)	kevin.brown@lam.com
Department	Deposition R&D
Location	Tualatin, OR
Cost Center	CC-4300 Deposition R&D
Priority	Low
Required By Date	2026-12-31
Estimated Budget	12000
Request Status	Cancelled
Current Phase	APPG Review
APPG Owner (email)	david.chen@lam.com
Approved Amount	(empty)
Submitted On	2026-08-25 11:00
Status Changed On	2026-09-11 08:00
Closed On	(empty)
Business Justification	Check adhesion of new low-k film
Technical Description	Tape test and 4-point bend
Expected Outcome	Adhesion > 5 J/m2
Rejection / Cancellation Reason	Auto-cancelled: no response for 30 days

Row 4 – TV-2026-00011 · Plasma etch uniformity (duplicate)	
TVID	TV-2026-00011
Request Title	Plasma etch uniformity (duplicate)
Request Type	New
Previous Request	(empty)
Recurrence Frequency	(empty)
Requester (email)	john.miller@lam.com
Department	Metrology
Location	Fremont, CA
Cost Center	CC-4200 Metrology
Priority	Medium
Required By Date	2026-12-15
Estimated Budget	40000
Request Status	Rejected
Current Phase	APPG Review
APPG Owner (email)	david.chen@lam.com
Approved Amount	(empty)
Submitted On	2026-09-01 10:00
Status Changed On	2026-09-03 12:00
Closed On	(empty)
Business Justification	Uniformity issue
Technical Description	Same as other request
Expected Outcome	Uniformity < 2%
Rejection / Cancellation Reason	Duplicate of TV-2026-00015 scope

Row 5 – TV-2026-00012 · EUV resist outgassing study	
TVID	TV-2026-00012
Request Title	EUV resist outgassing study
Request Type	New
Previous Request	(empty)
Recurrence Frequency	(empty)
Requester (email)	anna.white@lam.com
Department	Product Engineering
Location	Villach
Cost Center	CC-4300 Deposition R&D
Priority	High
Required By Date	2027-01-31
Estimated Budget	62000
Request Status	Pending Funding Approval
Current Phase	Funding Approval
APPG Owner (email)	david.chen@lam.com
Approved Amount	(empty)
Submitted On	2026-09-05 09:30
Status Changed On	2026-10-08 10:00
Closed On	(empty)
Business Justification	Customer qualification needed
Technical Description	Outgassing measurement at IMEC
Expected Outcome	Outgassing below limit
Rejection / Cancellation Reason	(empty)

Row 6 – TV-2026-00015 · Plasma etch uniformity test on 300mm wafers	
TVID	TV-2026-00015
Request Title	Plasma etch uniformity test on 300mm wafers
Request Type	New
Previous Request	(empty)
Recurrence Frequency	(empty)
Requester (email)	priya.sharma@lam.com
Department	Process Engineering
Location	Fremont, CA
Cost Center	CC-4100 Etch R&D
Priority	High
Required By Date	2026-12-15
Estimated Budget	42000
Request Status	In Execution
Current Phase	Execution
APPG Owner (email)	david.chen@lam.com
Approved Amount	39500
Submitted On	2026-10-12 10:05
Status Changed On	2026-11-10 08:30
Closed On	(empty)
Business Justification	Customer X reports 4% within-wafer non-uniformity on 3nm etch
Technical Description	25 wafers, 3 recipes, CD-SEM at 49 points
Expected Outcome	Uniformity below 2%
Rejection / Cancellation Reason	(empty)

Row 7 – TV-2026-00016 · Repeat chamber stress test with new liner	
TVID	TV-2026-00016
Request Title	Repeat chamber stress test with new liner
Request Type	Repeat
Previous Request	TV-2026-00003
Recurrence Frequency	(empty)
Requester (email)	priya.sharma@lam.com
Department	Process Engineering
Location	Fremont, CA
Cost Center	CC-4100 Etch R&D
Priority	Medium
Required By Date	2027-02-28
Estimated Budget	30000
Request Status	More Info Required
Current Phase	APPG Review
APPG Owner (email)	david.chen@lam.com
Approved Amount	(empty)
Submitted On	2026-10-01 09:00
Status Changed On	2026-10-06 09:00
Closed On	(empty)
Business Justification	Repeat of TV-2026-00003 with liner v2
Technical Description	Thermal cycling 500 cycles
Expected Outcome	No cracks
Rejection / Cancellation Reason	(empty)

Row 8 – TV-2026-00018 · Metrology tool correlation study	
TVID	TV-2026-00018
Request Title	Metrology tool correlation study
Request Type	One-time
Previous Request	(empty)
Recurrence Frequency	(empty)
Requester (email)	john.miller@lam.com
Department	Metrology
Location	Bangalore
Cost Center	CC-4200 Metrology
Priority	Low
Required By Date	2027-03-31
Estimated Budget	9000
Request Status	Submitted
Current Phase	APPG Review
APPG Owner (email)	(empty)
Approved Amount	(empty)
Submitted On	2026-10-09 15:00
Status Changed On	2026-10-09 15:00
Closed On	(empty)
Business Justification	Two CD-SEM tools give different results
Technical Description	Measure same 5 wafers on both tools
Expected Outcome	Offset < 0.5nm
Rejection / Cancellation Reason	(empty)

Sample data – feasibility_options.csv

Loads into table: Feasibility Option (tvr_feasibilityoption) · Import order: 8 of 10 · Rows: 5 · Columns: 9

Fake test data for DEV / TEST environments only – never load into PROD. People are given by e-mail and must exist in the environment first.

#	Option Name	TV Request	Research Institution	Capability Score	Quoted Cost	Timeline (weeks)	Constraints	Recommendation	Is Selected
1	Option A - Stanford Nanofab	TV-2026-00015	Stanford Nanofab Lab	5	39500	10	Lab free from Nov	Recommended	Yes
2	Option B - Fraunhofer	TV-2026-00015	Fraunhofer IPMS	4	35000	16	Shipping to Germany	Alternative	No
3	Option C - Internal Lab	TV-2026-00015	Internal Lab Fremont	3	22000	20	Tool booked till Jan	Not Recommended	No
4	Option A - IMEC	TV-2026-00012	IMEC	5	62000	12	EUV slot in Dec	Recommended	Yes
5	Option A - Internal Lab	TV-2026-00003	Internal Lab Fremont	4	28000	8	None	Recommended	Yes

Sample data – milestones.csv

Loads into table: Milestone (`tvr_milestone`) · Import order: 9 of 10 · Rows: 6 · Columns: 9

Fake test data for DEV / TEST environments only – never load into PROD. People are given by e-mail and must exist in the environment first.

#	Milestone Name	TV Request	Sequence	Planned Start	Planned End	Actual Start	Actual End	% Complete	Status
1	Wafer preparation	TV-2026-00015	1	2026-11-10	2026-11-24	2026-11-12	2026-11-26	100	Completed
2	Etch runs (3 recipes)	TV-2026-00015	2	2026-11-25	2026-12-15	2026-11-27	–	60	In Progress
3	CD-SEM measurement	TV-2026-00015	3	2026-12-16	2027-01-20	–	–	0	Not Started
4	Final report	TV-2026-00015	4	2027-01-21	2027-02-15	–	–	0	Not Started
5	Liner installation	TV-2026-00003	1	2026-08-20	2026-09-05	2026-08-21	2026-09-06	100	Completed
6	Thermal cycling	TV-2026-00003	2	2026-09-06	2026-10-31	2026-09-07	–	70	Delayed

Sample data – cost_entries.csv

Loads into table: Cost Entry (`tvr_costentry`) · Import order: 10 of 10 · Rows: 5 · Columns: 6

Fake test data for DEV / TEST environments only – never load into PROD. People are given by e-mail and must exist in the environment first.

#	Description	TV Request	Category	Amount	Cost Date	Invoice Reference
1	Lab usage fee - Nov	TV-2026-00015	Lab Fee	18000	2026-11-30	INV-88213
2	Wafers and consumables	TV-2026-00015	Material	5300	2026-11-15	INV-88190
3	Lab usage fee - Aug/Sep	TV-2026-00003	Lab Fee	14000	2026-09-30	INV-87002
4	Liner material	TV-2026-00003	Material	6000	2026-08-18	INV-86950
5	Quarterly check fee	TV-2026-00001	Lab Fee	14200	2026-06-20	INV-85010